AMC12 COMC Early/Mid-Band Problem Mastery Guide

Integrated Conversion Technique Recognition and Application

For AMC12 Problems 1–20 And COMC Part A - Part B

Advanced Mathematical Competition Training

2025 Edition - ENHANCED AND INTEGRATED

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Companion Document Reference

Important: Companion Document

This guide works in conjunction with **AMC12_COMC_Conversion_Techniques.tex** and

AMC12_COMC_Complete_Verification_Methods.tex. The conversion techniques document provides comprehensive coverage of 49 conversion techniques.

Verification companion overview (from AMC12_COMC_Complete_Verification_Methods.tex):

- **8 levels**: Instant Gut Check (1–3s), Minimal Sanity Check (5–10s), Targeted Spot Check (15–30s), Standard Verification (30–60s), Thorough (1–3m), Multi-Method (3–5m), Complete Mathematical (5–10m), Formal Proof-Level (10m+)
- **Guidance**: When to use each level by problem band and time budget

How to use both documents together:

1. Use this Mastery Guide for rapid pattern recognition and time management strategies
2. Refer to the Conversion Techniques document for detailed explanations and practice problems for each technique
3. Refer to the Complete Verification Methods document to choose the appropriate verification level and checklist
4. The success rates, time targets, and classifications are aligned across documents

Technique Numbering Convention: Throughout this guide, references like (T5), (T16), (T23) refer to specific technique numbers in the Conversion Techniques companion document. When a trap or mistake references a technique, consult that numbered section for detailed explanations and additional practice problems.

1 Strategic Problem Attack System

MASTER TIMELINE: Complete Problem Attack Strategy

Every Second Counts – Follow This Exact Timeline

Time	Phase	Primary Action	Success Metric
0–15s	CLASSIFY	Domain scan + pattern match	Domain identified
15–35s	EXECUTE	Apply domain-specific technique	Progress visible
35–50s	PIVOT	Switch technique if stuck	New approach started
50–65s	TEST	Verify via answer choices	Answer selected/narrowed
65–120s	BAILOUT	Emergency protocols	Skip decision made

Critical

Rule: Never exceed 2 minutes on any problem. Mark and move on.

WHO/WHEN Decision Framework

4–10 s

WHO: problem type the technique applies to (e.g., circle geometry, modular exponents)
WHEN: conditions/triggers for applying it (goal form, constraints, key signals, time window).

Quick Decision Protocol: If WHO+WHEN match within 6s → execute immediately. If unclear by 10s → proceed to structured classification.

WHO Problem Types Taxonomy

WHO Problem Types by Domain

Algebra (ALG): Linear/Absolute; Quadratics; Systems; Rational equations; Exponentials/Logs; Symmetric expressions; Polynomials (roots, remainders); Inequalities.
Geometry (GEO): Triangles (similarity/congruence, special centers); Circles (tangent/secant/-chords); Angle bisectors; Cevians (Ceva/Menelaus/Stewart/Mass Points); Coordinate geometry.
Counting/Combinatorics (CNT): Permutations/Combinations; Restrictions/"at least"; Stars and Bars (identical objects); Inclusion–Exclusion; Bijections; Graph models; Derangements.
Number Theory (NT): Divisibility/Primes/GCD/LCM; Modular arithmetic; CRT; Euler/FLT; Diophantine equations; Digit/base problems.
Probability (PROB): Sample spaces; Independence/Conditional; Linearity of Expectation; Geometric probability.
Sequences/Series (SEQ): Arithmetic/Geometric; Recurrences (characteristic equations); Telescoping; Power sums/Newton.

Keyword-to-WHO Recognition Map

Keyword → WHO (use for 0–4s domain scan)

Keyword patterns	**WHO (Domain/Type)**	**Notes (confidence/ambiguity)**
equation, solve for x, quadratic, roots, discriminant	ALG/Quadratic	Try T8-CompleteSquare, T16-Vieta, T33-RationalRoot
$xy + ax + by$, factorable bilinear	ALG/Bilinear	T5-SFFT first; integer targets favored
remainder, divisible by, last digit(s), base-b	NT/Digits–Mod	T19-ModArith, T9-BaseDigits, T20-FLT if prime
circle, tangent, secant, chords, power	GEO/Circle	T23-PoP; check cyclic/similarity
angle bisector, ratio of sides	GEO/Bisector	T10-ABT; Ceva/Mass Points as alternates
expected value/number, indicators	PROB/Expectation	T11-LOE; avoid independence assumption
"at least/at most", union/overlap	CNT/Union–Restrictions	T21-PIE; consider complement
identical objects into boxes	CNT/Distribution	TM1-StarsBars; enforce nonnegativity
recurrence with constant coefficients	SEQ/Recurrence	T13-CharEq; fit initial conditions
telescoping-looking terms, consecutive differences	SEQ/Telescoping	T29-Telescoping; partial fractions if set up
no clean synthetic relations, messy ratios	GEO/Coordinates	T2-CoordGeo; place symmetry at origin/axes
multiple congruences, coprime moduli	NT/CRT	T38-CRT; factor modulus
concurrent cevians, ratio chase	GEO/Ceva–Menelaus–Mass	T36-Ceva, T37-Menelaus, T35-MassPoints
power sums of roots, Newton sums	ALG/Power Sums	T27-PowerSum; connect to coefficients
functional equation cues $f(x+y)$, $f(ax)$, injective/surjective	ALG/FuncEq	T41-FuncEq; test $0, 1, -x$ and fixed points

WHEN Triggers Quick Index

WHEN to apply (conditions with key signals)

- **ALG**: Use T5-SFFT when bilinear structure visible; T16-Vieta when sums/products of roots queried; T33-RationalRoot when integer/rational roots plausible; T8-CompleteSquare when quadratic with square terms + linear.
- **GEO**: Use T23-PoP when tangent/secant/chords present; T10-ABT when a bisector partitions a side; pivot to T2-CoordGeo when synthetic relations stall by 25–30s.
- **CNT**: Use TM1-StarsBars for identical object distributions; T21-PIE on unions/overlaps; Complement when "at least" appears and direct counting is messy.
- **NT**: Use T19-ModArith for digits/last-digit/remainder; T20-FLT for prime modulus exponents; T38-CRT for systems of congruences.

- **PROB**: Use T11-LOE for expected counts without independence; switch to complement for "not/at least"; use conditional when sequential events given.
- **SEQ**: Use T13-CharEq for linear recurrences; T29-Telescoping when consecutive terms cancel; T27-PowerSum when symmetric polynomial data given.

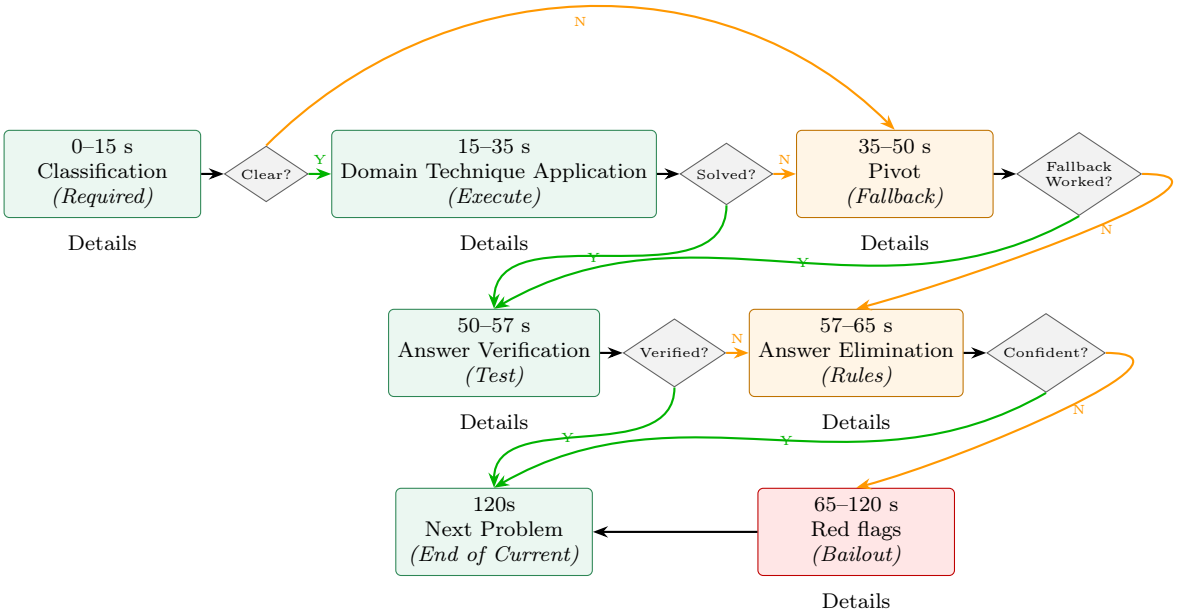
Fallback Technique Quicklist (WHO-anchored)

Pivot options after 30–35s without progress

- **ALG**: Swap T5-SFFT ↔ T16-Vieta; try T1-AlgSub to reduce degree; RRT (T33) on polynomials; bound/estimate to prune answers.
- **GEO**: Switch to T2-CoordGeo; if circle features present, test T23-PoP; for ratios, try T35-MassPoints or T10-ABT; Stewart (T31) for cevian length relations.
- **CNT**: Try complement; if multiple constraints, escalate to T21-PIE; small-case patterning to hypothesize formula.
- **NT**: Change modulus (2,3,4,5,7,8,9,11); use T38-CRT to combine; apply T20-FLT/T14-EulerPhi for exponents; invariants (T26) on move-process problems.
- **PROB**: Switch to T11-LOE; draw a tree for small depth; complement if "at least"/"at least one" appears.
- **SEQ**: Compute more terms; attempt T29-Telescoping; if composition/partitioning appears, attempt T25-GenFunc only if simple.

Timeline Legend with Decision Gates

Visual Time Pressure Indicators: Green = On Pace Yellow = Caution Red = Behind Schedule



Decision Gate Criteria:

- **Clear?** Domain and primary technique identified with confidence
  - YES → Move to Domain Lanes (Identify domains and techniques for execution)
  - NO → Move to Pivot Strategy (need different approach)
- **Solved?** Made meaningful progress with domain-specific technique
  - YES → Move to Answer Verification (solution ready for testing)
  - NO → Move to Pivot Strategy (need different approach)
- **Fallback Worked?** Pivot technique providing clear progress and manageable calculation
  - YES → Move to Answer Verification (made progress, ready to test solution)
  - NO → Move to Answer Elimination (systematic rule-out as backup approach)
- **Verified?** Answer verification successful during testing phase
  - YES → Proceed directly to Next Problem (skip elimination)
  - NO → Move to Answer Elimination for systematic rule-out
- **Confident?** Answer appears solid and constraints are satisfied
  - YES → Proceed to Next Problem (answer is reliable)
  - NO → Enter Red Flags protocols for deeper verification

1.1 15-Second Classification Phase

0–15 s Jump to Timeline **Success Rate: 85% for P1–P10 (Well-Prepared Students)**

Success Rate Variations by Preparation Level

Preparation Level	Classification (0-15s)	Execution (15-35s)	Overall Success
Advanced	90-95%	85-90%	80-85%
Well-Prepared	85%	75%	65-70%
Intermediate	70-75%	60-65%	45-50%
Beginner	50-60%	40-45%	25-30%

Note: Success rates apply to problems 1-10. For problems 11-20, reduce all percentages by 15-20%.

Three-Step Rapid Classification with Checkpoints

0–15 s **CRITICAL: Start timer NOW**
EXECUTE NOW: Start your timer. You have exactly **15s**.
Checkpoint Protocol: Mark progress at 4s, 10s, and 15s. If behind schedule, jump to next step.

1.1.1 Domain Scan (4s)

0–4 s **Target: 3s average, up to 6s for complex problems Visual Scan Protocol:** Circle key words/symbols and tag the domain:

- **Algebra** [*Keywords:* equations, expressions, logs, polynomials, functions, variables, solve for x, quadratics, systems, substitution, factoring, radicals, exponents, inequalities, Vieta's, SFFT, completing square]: Equations, expressions, exponents, logs, functions → Tag: ALG
- **Geometry** [*Keywords:* shapes, angles, coordinates, triangles, circles, area, perimeter, volume, similarity, congruence, tangent, chord, cyclic, inscribed, circumscribed, Power of Point, ABT, Stewart's, mass points]: Shapes, angles, circles, similarity, coordinates → Tag: GEO
- **Counting** [*Keywords:* counting, permutations, combinations, arrangements, selections, casework, inclusion-exclusion, complement, restrictions, "at least", "at most", indistinguishable, Stars and Bars]: "at least/at most", permutations → Tag: CNT
- **Number Theory** [*Keywords:* divisibility, primes, factors, multiples, modular, remainders, GCD, LCM, Diophantine, congruences, FLT, Euler, digits, base conversion, prime factorization]: Integers, modular arithmetic → Tag: NT
- **Probability** [*Keywords:* probability, expected value, outcomes, events, conditional, independence, mutually exclusive, geometric probability, random variables, distributions, "at least", "at most"]: Independence, conditional → Tag: PROB
- **Sequences/Series** [*Keywords:* sequences, series, patterns, arithmetic, geometric, recursive, telescoping, generating functions, limits, Fibonacci, summation, nth term, finite/infinite series]: Patterns, formulas, telescoping → Tag: SEQ

4s Checkpoint: Domain identified? If not, mark as the default ALG domain and continue.
Mixed-Domain Guidance: For MIXED domains: Identify the primary domain for initial approach, but prepare secondary domain techniques. Example: Counting problem with geometric constraint → start with counting, use geometry to establish bounds. Common mixed patterns:

- Algebra + Number Theory: Diophantine equations, modular polynomials
- Geometry + Counting: Grid paths, lattice points, symmetric configurations
- Probability + Geometry: Geometric probability, random points in regions
- Number Theory + Counting: Divisibility constraints, digit problems

1.1.2 Technique Selection + Setup (6s)

4–10 s **Target: 6s for pattern match Recognition Protocol:** Use the tiered Technique Recognition Cues in order: *Instant* (1–2s) then *Quick* (to 10s). Identify the first viable tactic using **WHO/WHEN** cues (WHERE folded into WHEN as key signals/conditions).

Workflow Checklist (6s):

- **Recognize (1–2s):** Try *Instant* cues first; if none, begin a brief *Quick* scan; pick the top-fit cue from Technique Recognition Cues.
- **Validate (1s):** Confirm WHEN conditions/prerequisites (e.g., positivity, integrality, concurrency); treat any key signals previously labeled as WHERE as part of WHEN.
- **Setup (2–3s):** Apply a minimal domain template:
  - **ALG:** Name substitution or invariant; isolate core expression; normalize scale/signs.
  - **GEO:** Choose coordinates or synthetic line; place symmetry at origin/centroid; define key lengths.
  - **CNT/PROB:** Define sample space; encode states; choose complement/indicators if simpler.
  - **NT:** Choose modulus/base; reduce to residue classes; note cycles/periods.
- **Checkpoint ($\leq$ 10s):** If no *Instant* match, complete a *Quick* scan. By 10s either a clear first equation/diagram emerges *or* pivot to the next step.
- **Answers Link:** If uncertain, glance at answers and apply Answer-Choice Intelligence to choose execution mode.

Classification Capture Checklist [0–15s]

Record the following before leaving 0–15s:

- Domain tag(s): primary (and secondary if mixed)
- Problem objective (in your own words)
- WHO (problem type)
- WHEN (triggers/conditions; key signals folded into WHEN)
- Candidate technique(s): exact short names (e.g., T5-SFFT)
- Answer-choice intelligence: format cues, execution mode, confidence (G/Y/R)

Where to Look: Use the *Instant* and *Quick* tables in Technique Recognition Cues. Maintain a single source of truth there; avoid duplicate lists here.

Recognition Tier Priority

Why Instant → Quick?

- *Instant* cues are cross-domain, low-setup triggers that fire in 1–2s.
- *Quick* cues require small setup and are domain-aware; use them only if no Instant cue fits.
- This order maximizes early wins and prevents over-setup during classification.

How to Apply

1) Check *Instant* cues (1–2s).
2) If none, scan *Quick* cues to complete the 4–10s window.
3) Confirm WHERE/WHEN constraints, then make the first move.

Example

Repeated chunk → T1-AlgSub (Instant). Circle with tangent/secant → T23-PoP (Instant). If the figure lacks direct relations but has symmetry, use T2-CoordGeo (Quick).

1.1.3 Answer-Choice Intelligence (5s)

10–15 s **Target: 3s scan, 2s decision Answer Analysis Protocol:** Use the answers to select execution mode and validate approach:

- Sortable numerical choices → prepare constraint elimination; test middle only after establishing monotonicity, see 50-Second Answer Testing
- Radicals/π present → Exact geometry; avoid rough rounding
- Fractions/ratios → Set a proportion; clear denominators
- Probabilities → Eliminate < 0 or > 1; prefer complements
- Expressions in parameters → Manipulate algebraically; only plug if invariant

Decision Point at 15s – Critical Fork:

- **GREEN PATH** (confidence $> 80\%$): Proceed with identified technique → Domain Lanes
- **YELLOW PATH** (confidence 50–80%): Choose simpler option, set 30s bailout timer
- **RED PATH** (confidence $< 50\%$): Skip immediately, mark with domain tag for return

Scratch Paper Protocol:

- Mark: G (execute now), Y (review at 35s), R (skip)
- Add 1–2 word technique cue (e.g., "SFFT", "PoP", "Vieta")
- Note time invested: 15s baseline

Bailout Trigger: If unclear by 15s, jump to 35-Second Pivot.

Confidence Calibration Box

At 15s: Quick Self-Assessment

Rate Your Confidence (1-10):

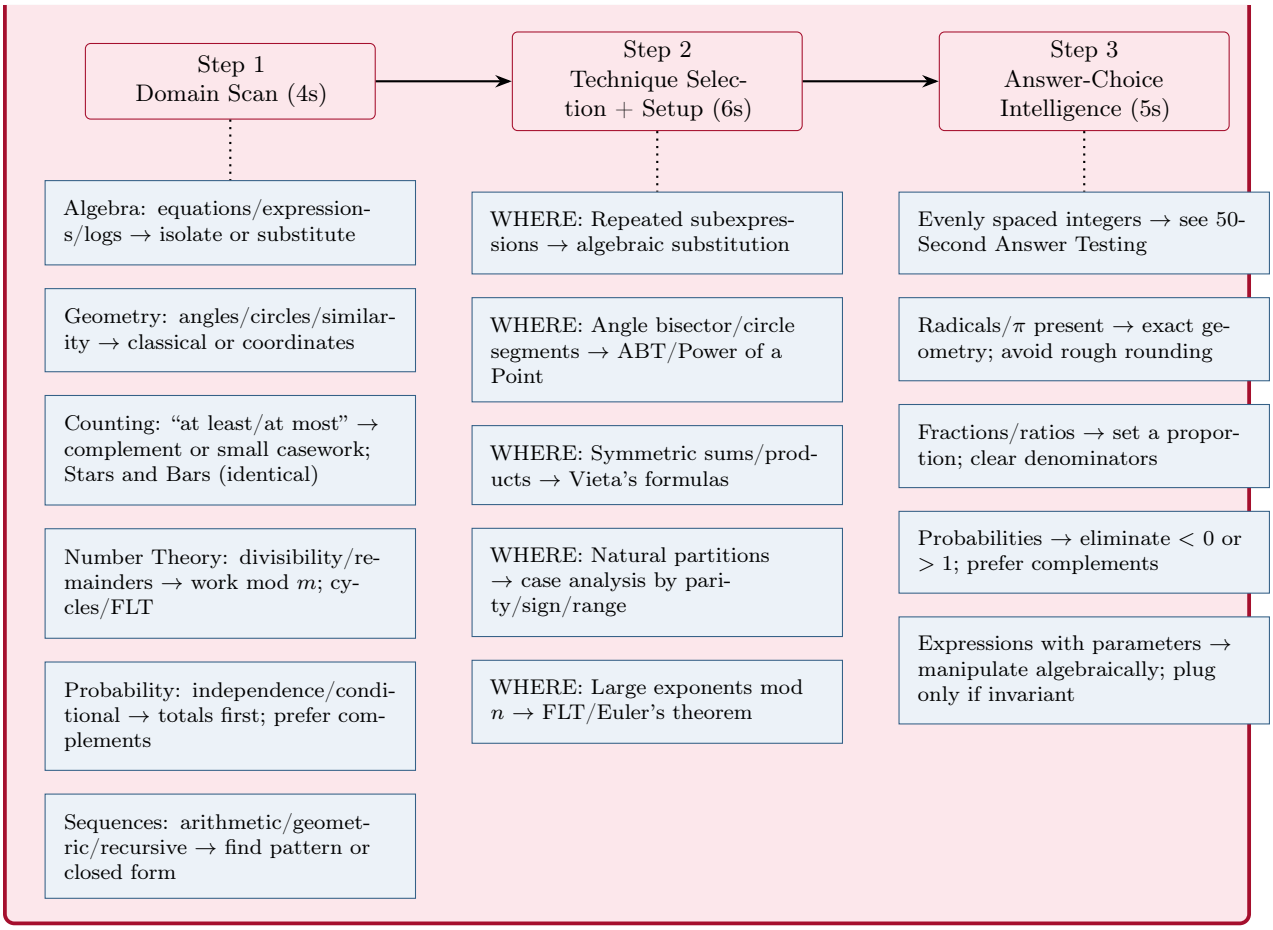
- **8-10**: Strong confidence → Mark G, execute immediately
- **5-7**: Moderate confidence → Mark Y, proceed with 35s checkpoint
- **3-4**: Low confidence → Mark Y, prepare to pivot at 35s
- **1-2**: Very low confidence → Mark R, skip immediately

Calibration Tips:

- If you can't identify the domain within 5s, reduce confidence by 2 points
- If answer choices seem unusual or inconsistent, reduce confidence by 1 point
- If you've solved similar problems successfully before, increase confidence by 1 point

Visual Workflow: 15-Second Classification

0–15 s


Step 1
Domain Scan (4s)
Step 2
Technique Selection + Setup (6s)
Step 3
Answer-Choice Intelligence (5s)
Algebra: equations/expressions/logs → isolate or substitute
Geometry: angles/circles/similarity → classical or coordinates
Counting: "at least/at most" → complement or small casework; Stars and Bars (identical)
Number Theory: divisibility/remainders → work mod m; cycles/FLT
Probability: independence/conditional → totals first; prefer complements
Sequences: arithmetic/geometric/recursive → find pattern or closed form
WHERE: Repeated subexpressions → algebraic substitution
WHERE: Angle bisector/circle segments → ABT/Power of a Point
WHERE: Symmetric sums/products → Vieta's formulas
WHERE: Natural partitions → case analysis by parity/sign/range
WHERE: Large exponents mod n → FLT/Euler's theorem
Evenly spaced integers → see 50-Second Answer Testing
Radicals/π present → exact geometry; avoid rough rounding
Fractions/ratios → set a proportion; clear denominators
Probabilities → eliminate < 0 or > 1; prefer complements
Expressions with parameters → manipulate algebraically; plug only if invariant

1.2 20-Second Execution Phase

15–35 s Jump to Timeline **Success Rate: 75% for classified problems**

*Timeline: **15–20s** immediate → **20–25s** structure → **25–30s** core technique → **30–35s** lane fallback probe → **35s** **hard pivot** if no short, deterministic finish is visible.*

QUICK DECISION CARD - 15 Second Checkpoint

STOP - ASSESS - DECIDE

Immediate Classification Decision:

Situation at 15s	Action	Reference
Domain clear + technique obvious	Execute immediately	→ Mini-Checklist
Domain clear, technique uncertain	Check patterns	→ Visual Workflow
Multiple domains visible	ID primary, note secondary	→ Primary Lane
Domain completely unclear	Default to Algebra	→ Algebra Lane
Confidence < 50%	Mark YELLOW	→ Prepare pivot

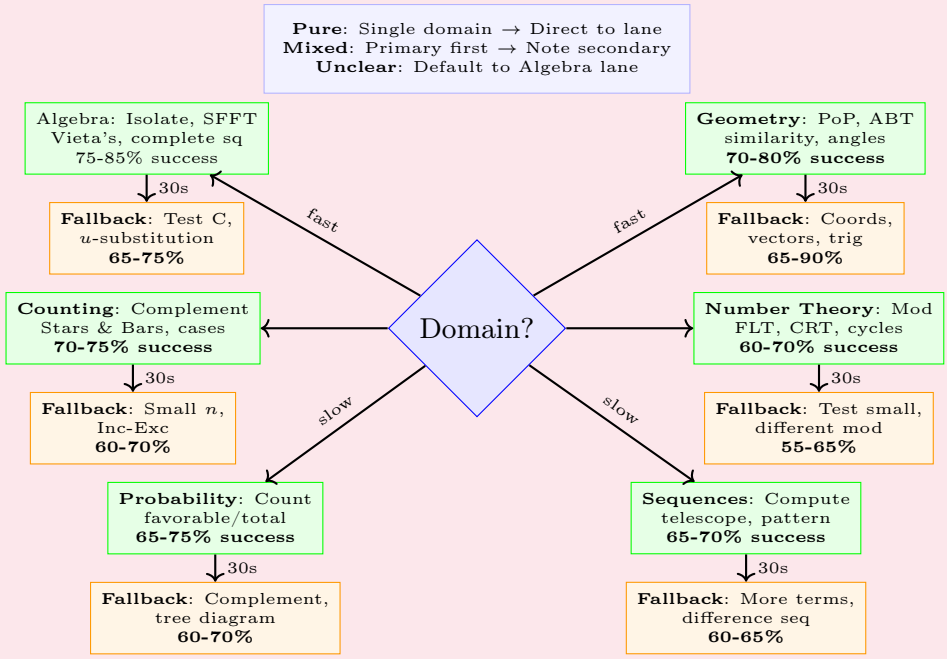
Time Investment Matrix by Problem Number:

Problem	Confidence	Time Budget	Strategy	Bailout
1–5	80%+	65s full	Complete execution	120s hard stop
1–5	50-79%	45s	Primary + quick verify	90s
1–5	<50%	30s max	Quick attempt only	Skip early
6–10	80%+	120s full	Full method + verify	180s hard stop
6–10	50-79%	90s	Primary + fallback	150s
6–10	<50%	60s	One attempt + skip	Move on
11–15	80%+	180s	Multiple techniques	210s hard stop
11–15	50-79%	120s	Primary + one fallback	180s
11–15	<50%	Skip	Mark for later	–
16–20	70%+	240s	Full exploration	270s hard stop
16–20	<70%	Skip	Only if time remains	–

Personal Success Rate Reference:
Compare your domain success rates (track in practice) to decide time investment

Visual Workflow: Classification to Action

15–35 s



Speed Guide: Green = Usually <20s | Orange = 20-30s | Red = Often 30s+
Critical Times: 15s (classify) → 25s (assess) → 30s (pivot) → 35s (complete)

Quick Decision Protocol at 15s:

- ☐ Domain clear + technique obvious → Go to Mini-Checklist below
- ☐ Domain unclear → Use this workflow to identify
- ☐ Mixed domain → Start with primary, prepare secondary
- ☐ Confidence < 50% → Mark YELLOW, prepare early pivot

Domain Lane Mini-Checklists with Timing Milestones

15–35 s **Execute ONE lane based on classification**
Reference: See Visual Workflow above for domain selection and fallback options

1.2.1 Algebra Lane (20s)

15–35 s Tag: ALG *Typical payoff: Primary high | Fallback moderate*

- ▶ **[15–20s] CRITICAL - Immediate action**:
  - Isolate variable or substitute repeated chunk
  - For xy terms with linear parts: use SFFT (T5)
  - For quadratics: complete square or apply Vieta's (T16)
- ○ **[20–25s] Pattern recognition**:
  - Hidden quadratic? Try $u = x^2$ or $u = x + \frac{1}{x}$
  - Symmetric expressions? Use $s = x + y$, $p = xy$
  - Absolute value present? Set breakpoints for piecewise
- **[25s CHECKPOINT]** Progress Assessment:
  - ✓ **Visible progress**: Continue current approach
  - × **No progress**: Prepare fallback strategy now
- ▷ **[25–30s] Core technique execution**:
  - Bilinear $xy + ax + by = c \to (x + \alpha)(y + \beta) = \gamma$
  - Check discriminant $b^2 - 4ac \geq 0$ for real solutions
  - Quick test: Try $x \in \{0, 1, -1, 2\}$ before complex algebra
  - Univariate integer polynomial? Try Rational Root Test and quick factor scan.
  - Symmetric or nearly symmetric? Try $u = x + y$, $v = xy$ reduction.
- ↻ **[30–35s] Fallback execution**:
  - Primary: u-substitution for complex expressions
  - Secondary: If choices are numeric and the target is monotone in a single parameter, test a *median-magnitude* choice as a *scale check* only (do not select by position).
  - Emergency: Graph or bound the expression
  - *Hard cut at 35s: if no clean, near-term path (<10s arithmetic) is visible, trigger 35-Second Pivot Strategy.*
- **[!] Common traps**: Sign errors; extraneous from squaring; domain restrictions
- **Most Common Pattern:** Hidden quadratic → u-substitution *(very common)*

1.2.2 Geometry Lane (20s)

15–35 s Tag: GEO *Typical payoff: Primary high | Fallback moderate*

- ▶ **[15–20s] CRITICAL - Immediate action**:
  - Circle present? → Power of a Point (T23)
  - Angle bisector? → ABT (T10)
  - Otherwise: Mark all equal angles/sides
- ○ **[20–25s] Relationship scan**:
  - Check similarity (AA/SAS/SSS)
  - Cyclic quadrilateral? (opposite angles sum to 180°)
  - Pythagorean triples: (3,4,5), (5,12,13), (8,15,17)
- **[25s CHECKPOINT]** Synthetic Progress:
  - ✓ **Relations found**: Continue synthetic
  - × **Stuck**: Set up coordinates immediately
- ▷ **[25–30s] Synthetic techniques**:
  - Stewart's theorem for cevians

  - Mass points for ratios
  - Special centers: circumcenter, incenter, centroid
- ↻ **[30–35s] Coordinate fallback**:
  - Place origin at symmetry point
  - Use simple integer coordinates
  - Work with d^2 to avoid radicals: $(x_2 - x_1)^2 + (y_2 - y_1)^2$
  - *Hard cut at 35s: if no clean, near-term path (<10s arithmetic) is visible, trigger 35-Second Pivot Strategy.*
- **[!] Common traps**: Assuming perpendicularity; scaling errors (length vs area)
- **Most Common Pattern:** Circle problem → Power of a Point *(very common)*

1.2.3 Counting Lane (20s)

15–35 s Tag: CNT *Typical payoff: Primary high | Fallback moderate*

- ▶ **[15–20s] CRITICAL - Classification**:
  - Order matters? Yes → Permutation, No → Combination
  - "At least" present? → Use complement
  - Identical objects? → Stars and Bars
  - Stars & Bars assumes nonnegative integers. For lower bounds $x_i \geq L_i$, substitute $x_i' = x_i - L_i$. For upper bounds, use complement or a 1–2 level PIE.
- ○ **[20–25s] Structure recognition**:
  - Multiple constraints? → Inclusion-Exclusion
  - Circular arrangement? → Fix one, count rest
  - Derangements? $D_n = n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}$
- **[25s CHECKPOINT]** Method Clear:
  - ✓ **Formula identified**: Execute calculation
  - × **Uncertain**: Start small cases immediately
- ▷ **[25–30s] Formula execution**:
  - $C(n, r) = \frac{n!}{r!(n-r)!}$
  - Stars and Bars: $C(n + k - 1, k - 1)$ for n identical into k bins
  - Check magnitude reasonable for problem number
  - Overlaps present? Apply PIE with at most 2 inclusion levels in this window.
- ↻ **[30–35s] Small cases fallback**:
  - Compute $n = 1, 2, 3$ to find pattern
  - Look for recursion relation
  - Break into disjoint subcases
  - *Hard cut at 35s: if no clean, near-term path (<10s arithmetic) is visible, trigger 35-Second Pivot Strategy.*
- **[!] Common traps**: Off-by-one errors; overcounting symmetries; missing cases
- **Most Common Pattern:** "At least" problem → Complement *(frequent)*

1.2.4 Number Theory Lane (20s)

15–35 s Tag: NT *Typical payoff: Primary moderate | Fallback low-moderate*

- ▶ **[15–20s] CRITICAL - Immediate action**:
  - Reduce modulo small primes (2, 3, 5, 7)
  - Powers? Apply Fermat's Little Theorem
  - Look for cycles in remainders
  - Quick divisibility checks: gcd via Euclid; mod-11 test; parity/mod-4/mod-8 for squares
- ○ **[20–25s] Pattern identification**:
  - Diophantine equation? Factor using SFFT

  - Multiple congruences? Set up for CRT
  - Digit problem? Use base representation
- **[25s CHECKPOINT]** Pattern Status:
  - ✓ **Cycle found**: Complete pattern
  - × **No pattern**: Test small values
- ▷ **[25–30s] Core techniques**:
  - FLT: $a^{p-1} \equiv 1 \pmod{p}$ if $\gcd(a, p) = 1$
  - CRT for system of congruences
  - Digital root for divisibility by 9
- ↻ **[30–35s] Testing fallback**:
  - Compute first 5-6 terms for pattern
  - Try different modulus (especially prime)
  - Check quadratic residues if squares involved
  - *Hard cut at 35s: if no clean, near-term path (<10s arithmetic) is visible, trigger 35-Second Pivot Strategy.*
- **[!] Common traps**: Can't cancel unless $\gcd(a, n) = 1$; 1 is not prime
- **Most Common Pattern:** Modular powers → Find cycle *(frequent)*

1.2.5 Probability Lane (20s)

15–35 s Tag: PROB *Typical payoff: Primary moderate-high | Fallback moderate*

- ▶ **[15–20s] CRITICAL - Setup**:
  - Count favorable outcomes and total outcomes
  - Check if complement is simpler
  - Verify all probabilities in [0,1]
- ○ **[20–25s] Structure analysis**:
  - Independent events? Multiply probabilities
  - Conditional? $P(A|B) = \frac{P(A \cap B)}{P(B)}$
  - Expected value? Use linearity: $E[X + Y] = E[X] + E[Y]$
  - Confirm the sample space is uniform; if not, weight appropriately.
- **[25s CHECKPOINT]** Counting Correct:
  - ✓ **Counts verified**: Simplify fraction
  - × **Uncertain**: Try complement approach
- ▷ **[25–30s] Calculation**:
  - Simplify fraction to lowest terms
  - For expectation: Use indicators if complex
  - Geometric probability: Use area/volume ratios
  - Use linearity of expectation (works without independence).
- ↻ **[30–35s] Complement fallback**:
  - $P(A) = 1 - P(A^c)$ often simpler
  - Draw probability tree for visualization
  - Break into disjoint events
  - *Hard cut at 35s: if no clean, near-term path (<10s arithmetic) is visible, trigger 35-Second Pivot Strategy.*
- **[!] Common traps**: Double-counting; not normalizing; assuming independence
- **Most Common Pattern:** Complex event → Use complement *(very common)*

1.2.6 Sequences/Series Lane (20s)

15–35 s Tag: SEQ *Typical payoff: Primary moderate-high | Fallback moderate*

- ▶ **[15–20s] CRITICAL - Compute terms**:
  - Calculate first 3-4 terms explicitly
  - Check arithmetic (constant difference)
  - Check geometric (constant ratio)
- ○ **[20–25s] Pattern search**:
  - Telescoping? Look for cancellation
  - Recursive? Find closed form
  - Periodic? Identify cycle length
  - Constant 2nd differences ⇒ quadratic (3rd ⇒ cubic); solve by finite differences.
- **[25s CHECKPOINT]** Pattern Found:
  - ✓ **Formula clear**: Apply and compute
  - × **No pattern**: Compute more terms
- ▷ **[25–30s] Formula application**:
  - Arithmetic: $S_n = \frac{n(a_1+a_n)}{2}$
  - Geometric: $S_n = a_1 \cdot \frac{1-r^n}{1-r}$ if $r \neq 1$
  - Telescoping: Write expanded form
  - For rational terms, try partial fractions to expose telescoping.
- ↻ **[30–35s] Extended computation**:
  - Compute 5-7 terms for pattern
  - Try difference sequence $b_n = a_{n+1} - a_n$
  - Check for hidden arithmetic/geometric
  - *Hard cut at 35s: if no clean, near-term path (<10s arithmetic) is visible, trigger 35-Second Pivot Strategy.*
- **[!] Common traps**: Off-by-one in indices; missing initial terms; wrong formula
- **Most Common Pattern:** Hidden telescoping → Partial fractions *(frequent)*

1.3 15-Second Pivot Phase

35–50 s Jump to Timeline

*This is a cross-domain pivot that **supersedes** the domain lanes after 35s. Continue in-lane past 35s only if a short, deterministic finish (< 10s of arithmetic) is already in flight.*

35-Second Pivot Strategy: Detailed Actions (35-50s)

35–50 s

Overall Constraints:
- **10-second promise:** pick exactly one light method; if no traction in $\leq$ 10s, bail.
- **No heavy lifts:** avoid new variables, multi-branch casework, or long expansions after 35s.

1.3.1 Decision to Pivot (2s)

35–37 s

Trigger Conditions:
- Pattern recognition failed or unclear
- Primary technique not working
- Calculation becoming too complex
- Time pressure mounting
- No meaningful progress visible

Decision Process:
- **Progress Check**: Are you advancing toward solution?
- If making progress $\rightarrow$ Continue with current approach
- If stuck $\rightarrow$ Enter pivot mode and select from menu below

Pivot Menu (pick one in $\leq$ 2s):
1. **Monotonicity $\Rightarrow$ bound a choice**: sign/range check, then test a median-magnitude choice.
2. **Parity / Mod**: reduce to impossibility/uniqueness quickly.
3. **Complement / Symmetry**: count the easy part or exploit invariance.
4. **Coordinate / Similar triangles** (geo): drop radicals via d^2, ratio-chase.
5. **Indicators + LOE** (prob): expected count beats full distribution.
6. **RRT / Vieta** (alg): integer roots or sum/product short-cuts.

1.3.2 Apply Secondary Technique (10s)

37–47 s

For Algebra Problems:
- Try substitution: set u to a *repeated/complicated subexpression* (e.g., $x + \frac{1}{x}$, $x^2 + xy + y^2$).
- If choices are numeric and the target is *monotone* in a single parameter, probe a *median-magnitude* choice as a *scale check only* (do not prefer "C" by position).
- Try simple values: $x = 1, 2, 3$
- Graph both sides if possible

For Geometry Problems:
- Switch to coordinates immediately
- Place origin at symmetry point
- Use simple integer coordinates
- Apply distance/slope formulas

For Counting Problems:
- Try small cases ($n = 1, 2, 3$)
- Use complement approach

- Break into simpler subcases
- Look for pattern in first few values

For Number Theory Problems:

- Tiny cases to spot cycles; then test modulus patterns
- Quick mods: $2, 3, 4, 5, 7, 8, 9, 11$; squares $\bmod 4, 8$; digital sums $\bmod 9, 11$
- Valuations v_2, v_5 for trailing zeros / divisibility
- Use orders/cycles (e.g., Fermat/Euler) only if it shortens computation

For Probability Problems:

- Use complement: $P(A) = 1 - P(\text{not } A)$
- Simplify to smaller case
- Draw probability tree
- Check if answers sum to 1
- Use indicators: let $X = \sum_i \mathbf{1}_{E_i}$, then $\mathbb{E}[X] = \sum_i \mathbb{P}(E_i)$ (no independence needed).
- Confirm sample space uniformity; if not, weight appropriately.

For Sequences Problems:

- Compute more terms (first 3-4)
- Check for telescoping pattern
- Look for recursion relation
- Test if arithmetic/geometric

Emergency High-Priority Techniques (if domain-specific fails):

- **SFFT**: For $xy + ax + by = c$ problems
- **Vieta's**: For polynomial roots ($x_1 + x_2 = -b/a$, $x_1x_2 = c/a$)
- **Power of a Point**: For circle chord/tangent problems
- **Angle Bisector Theorem**: $BD/DC = AB/AC$
- **Stars & Bars**: $\binom{n+k-1}{k-1}$ for nonnegative integers. Lower bounds: shift $x_i' = x_i - L_i$. Upper bounds: use complement or 1–2 level PIE.
- **Linearity of Expectation**: $E[\sum X_i] = \sum E[X_i]$
- **Inclusion–Exclusion**: $|A \cup B| = |A| + |B| - |A \cap B|$
- **AM–GM/Cauchy–Schwarz**: quick bounds for symmetric sums; prefer AM–GM if it tightens immediately.

1.3.3 Quick Verification or Setup for Next Phase (3s)

47–50 s

Universal Fallback Methods:

- Apply Constraints First: Eliminate using bounds/parity/mod before choice testing
- Magnitude Bounds: Eliminate answers differing by 10× or more
- Parity Check: Eliminate based on even/odd
- Unit Analysis: Check dimensions match
- Bound and Eliminate: Get rough bounds to eliminate 2-3 answers

Quick Approximations to Use:

- $\pi \approx 3.14$ (or 3 for very rough)
- $\sqrt{2} \approx 1.41$
- $\sqrt{3} \approx 1.73$
- $e \approx 2.72$
- Golden ratio $\phi \approx 1.62$

Exit Criteria at 50s:

- **Solved**: move to Answer Verification only (no extra work).
- **Partial but fragile**: jump to Answer Verification and confirm with a single plug-back.
- **No path**: begin **Choice Elimination** using sign/range/parity/mod; test at most *one* candidate.

Key Success Factors:

- Don't hesitate - commit to pivot quickly at 35s

- Choose simpler approach over complex
- Prioritize answer testing over complete solution
- Use mental math and approximations
- Keep track of eliminated choices

This 15-second window is critical for course correction and often determines whether you'll solve the problem or need to make an educated guess.
Full technique details: Technique Recognition Cues.

1.4 7-Second Answer Verification Phase

50–57 s Jump to Timeline

Answer Verification Strategy – Level 4 Protocol (7s budget)

50–57 s **Mathematical verification using Level 4 Standard Protocol**

When to Enter This Phase:

- Arrived via "Fallback Worked? → Yes" from Pivot phase
- Algebraic manipulation exceeding 15+ steps without simplification
- Multiple approaches attempted, no clear winner
- Answer obtained but seems suspicious

Stop Rules (enforced throughout):

- **One plug-back only, one alternate view only**
- **No new variables, expansions, or multi-branch casework after 50s**
- **Hard exit at 57s** → immediately enter Answer Elimination if unresolved

1.4.1 Plug-back & Constraints (one pass, 2s)

50–52 s

Level 4 Standard Verification:

- **Direct substitution**: Plug answer back into *original* statement (not squared/cleared forms)
- **Domain check**: Verify denominators $\neq 0$, radicals/logs valid, ordering/distinctness constraints
- **Extraneous roots**: Check if squaring, absolute values, or rationalizing occurred
- **Arithmetic verification**: Re-check one key step from your main solution path

Domain-Specific Quick Checks (pick what applies):

- **Combinatorics/Number Theory**: Answer must be integer; factorial/binomial inputs non-negative; residues/parity match
- **Probability**: Result in $[0, 1]$; probabilities sum to 1; expectation in feasible range
- **Geometry**: Triangle inequality; use d^2 for consistency; similar-triangle ratios; scale sanity
- **Sequences/Series**: Closed form reproduces first 2-3 terms; telescoping endpoints correct

1.4.2 Invariants & One Alternate View (3s)

52–55 s

Invariants Sweep (Level 2 Minimal):

- **Sign, range, parity**: Does answer match expected properties?
- **Modular arithmetic**: Check residues mod 2, 3, 4, 5 if relevant
- **Scale/homogeneity**: Length $\to k$, area $\to k^2$; dimensional consistency
- **Simple bounds**: Compare to rough estimate from pivot phase

One Micro Alternative (pick one, no new algebra):

- Tiny-case consistency ($n = 1, 2$ reproduce answer pattern)
- Quick monotonicity glance (if applicable)
- Single ratio/similarity check (geometry)

- Reasonableness vs bound/estimate

1.4.3 Choice Sanity & Commit (2s)

55–57 s

Guarded Choice Probing (only if justified):

- **Median-magnitude test**: Only after establishing *monotonicity* or clear unimodality
- **Extreme boundary test**: Only if target is provably *monotone* or extreme-attaining
- **No "Test (C)" bias**: Position-based preferences are unreliable

Rapid Estimation (use only when choices well-separated):

- Magnitude bounds: Eliminate answers differing by 10× or more
- Quick approximations: $\pi \approx 3$, $\sqrt{2} \approx 1.4$, $e \approx 2.7$ (risk of rounding error)

Exit Decision at 57s:

- **High confidence**: Select answer and move on (no extra work)
- **Likely candidate**: Commit if satisfies invariants; otherwise → Answer Elimination
- **Conflicting/uncertain**: Immediately → Answer Elimination (do not extend verification)

This 7-second verification window aligns with Level 4 Standard Protocol: plug-back/domain first, invariants second, choice probing last with proper guardrails.

1.5 Answer Elimination Phase [AMC-ONLY, 8s]

57–65 s Jump to Timeline

Two-Stage Constraint-First Elimination – Level 4 Protocol

57–65 s **Disciplined, evidence-based elimination with hard time caps**

When to Enter This Phase:

- Arrived via "Fallback Worked? → No" from Pivot phase
- Completed verification phase without definitive answer
- Need to narrow down remaining choices using objective constraints
- Less than 15s remaining on problem

CRITICAL GUARDRAILS (enforced throughout):

- **No new variables, expansions, or multi-branch casework** after 57s
- **Cap:** $\leq$3 independent constraints total (2 standard + optional 1 if $\geq$3 remain) and $\leq$1 probe
- **Approximation only if choices well-separated** (>10-15% gaps)
- **Never rely on choice position without monotonicity/convexity evidence**

1.5.1 Stage A: Constraint Sweep (4s)

57–61 s **Two independent constraints; optional 3rd only if $\geq$3 remain and $<$61s**

Apply **TWO** independent, high-yield constraints you already derived. Cross out incompatible choices immediately:

Prioritize orthogonality: pair different axes (e.g., sign/range + modular/type). Avoid redundant checks (e.g., mod 2 after mod 4).

Primary Constraint Categories:

- **Sign/Range**: Answer must be positive/negative, within bounds (a, b)
- **Parity/Modular**: Only if answer is integer/rational; use mod 2,3,4,5,7,8,9,11
- **Scale/Bounds**: AM-GM/Cauchy quick bounds; homogeneity scaling ($k \to k^2$ for areas)
- **Form/Type**: Must be integer-only, simplified radical, rational vs irrational

Domain-Specific Elimination Menu (use only what applies):

- **Algebra**: Sign/range from discriminant or AM-GM; Rational Root Test for integer/fraction choices; symmetry forms ($x \leftrightarrow \frac{1}{x}$ implies $x + \frac{1}{x}$ structure)

- **Number Theory**: Residues mod 2,3,4,5,7,8,9,11; quadratic residue screens ($\square$ mod $4, 8$); digit sum/last digit; quick v_2, v_5 valuations
- **Counting**: Must be non-negative integer; stars & bars constraints; compare growth rates (binomial vs polynomial)
- **Probability/Expectation**: Value in $[0, 1]$; complements sum to 1; expectation within feasible range
- **Geometry**: Triangle inequality; similar-triangle ratio consistency; prefer d^2 arithmetic; scaling sanity (length vs area vs volume)
- **Sequences/Series**: Candidate formula reproduces initial 2-3 terms; telescoping endpoints consistent; degree matches differences pattern

Stage A Goal: Reduce to 1–2 candidates. If a single candidate remains, commit and skip Stage B. If still ≥ 3 choices, optionally apply ONE additional independent constraint only if you remain under 61s; otherwise proceed to Stage B.

1.5.2 Stage B: Single Probe Tie-Break (4s)

61–65 s **Exactly one action to resolve ties**

Choose **exactly ONE** of the following actions:

- **Monotonicity Probe**: If monotonicity/convexity established, test a **median-magnitude** choice as scale check only
- **Special Value Check**: Plug a permitted special value (endpoints, $n = 1$, etc.) that separates remaining forms
- **Single Plug-Back**: Verify sign of key term for narrowest candidate (**no new algebra**)

Then COMMIT. If still tied, pick candidate satisfying **more independent constraints** (not letter position).

1.5.3 Final Decision Protocol

65 s **Hard deadline - no extensions**

- **CONFIDENT** (eliminated 3+ choices): Select best remaining candidate
- **UNCERTAIN** (2–3 choices remain): Pick candidate satisfying most constraints *(AMC EV > blank; commit)*
- **NO PROGRESS**:
  - 4 choices remain (AMC): EV $\approx$ blank; prefer **blank** unless one candidate weakly dominates
  - 5 choices remain (AMC): **Leave blank**

COMC Adaptation Note: Use the same **constraint sweep** strictly as a final sanity screen for a *computed* numeric candidate. Skip letter-based tie-breaks entirely; do not fabricate a number from constraints alone. If still uncertain, defer and return later rather than forcing a guess.

This constraint-first, two-stage approach ensures systematic, evidence-based elimination while preventing time overruns from low-yield choice probing.

1.6 55-Second Bailout Phase (65–120s)

Any Time Jump to Timeline **DANGER DETECTORS**

1.6.1 Global Red Flag Detectors (Trigger at Any Time)

10–120 s **Early warning system - can fire immediately**

High-Frequency Danger Patterns with Light Mitigations

Critical Detection → Light Action (≤ 10s) → Skip if unclear

(Tn) refers to "Technique n" in companion AMC12_ COMC_ Conversion_ Techniques document

Immediate Red Flags (10-20s detection):

1. **"Seems too easy"** [P8–12]:
   Light action (≤10s): Check $n = 0$, negative values, one boundary case
   Heavy trap signal: If still feels trivial → likely sophisticated trap → **SKIP**
2. **"Multiple interpretations"** [Ambiguous wording]:
   Light action (≤10s): Re-read key phrases once, sketch diagram
   Still ambiguous: → **SKIP** (don't guess interpretation under pressure)
3. **"All answers look similar"** [Differ by factor/sign only]:
   Light action (≤10s): Check one calculation step, verify signs
   Still unclear: Use constraint elimination → best remaining choice
4. **"All answers are close decimals"**:
   Light action (≤10s): Use exact bounds/parity/mod instead of estimation
   Avoid: Crude approximation traps → use objective constraints only

Structure-Based Red Flags (≤10s, ≤2 cases):

5. **"Nested floors/ceilings or absolute values everywhere"**:
   Light action: Try ≤2 cases (e.g., integer thresholds); ≤10s total
   3+ cases needed: → **SKIP**
6. **"PIE looks unavoidable and deep"**:
   Light action: Allow ≤2 inclusion levels only; ≤10s setup
   3+ levels needed: → **SKIP**
7. **"CRT with multiple non-coprime moduli"**:
   Light action: Try single small modulus to eliminate choices; ≤10s
   Full CRT needed: → **SKIP**
8. **"High-degree polynomial, no obvious RRT hits"**:
   Light action: Test ±1, ±2 only (T33-RationalRoot); ≤10s
   No factors found: → **SKIP**
9. **"Geometry with stacked theorems"** [Ceva/Menelaus/Mass Points/Stewart]:
   Light action: Try coordinates (T2) or one similar-ratio glance; use d^2 not d
   Multiple theorems needed: → **SKIP**
10. **"Probability with tangled dependencies"**:
    Light action: Try indicators + LOE (T11) for expected count; ≤10s
    Complex tree needed: → **SKIP**
11. **"Functional equation with no natural anchor"**:
    Light action: Test one anchor ($x = 0, 1, -1$) or symmetry check; ≤10s
    No traction: → **SKIP**
12. **Sequence with characteristic polynomial ≥ 3**:
    Light action: Compute 3 terms, try telescoping (T29) sniff; ≤10s
    No pattern: → **SKIP**

Emergency Decision Flow:

1. **Detect** red flag pattern
2. **One light pivot** (≤10s, ≤2 cases, no new variables/long expansions)
3. **Still unclear?** → Follow Skip Decision Framework below

1.6.2 65-120s Bailout Protocol

65–120 s **DANGER ZONE** - Maximum 2 minutes per problem

Late-Phase Emergency Protocol

When to Enter: Any problem still unresolved at 65s + major technique failure

Revised Emergency Mitigations (≤10s, bounded):

- **"No clear path"** → Try Coordinate Conversion (T2-CoordGeo) only (**not** Case Analysis T3-CaseAnalysis)
- **"Ugly numbers"** → Telescoping (T29-Telescoping), partial fractions if already set up (**demote** T25-GenFunc to study-time)
- **"Technique failing"** → One technique swap: SFFT(T5-SFFT)↔Vieta's(T16-Vieta), PoP(T23-PoP)↔Coordinates(T2-CoordGeo)

Skip Decision Framework [Strict Numeric Gates]:

- **2 Minutes Hard Stop**: No progress after 120s → SKIP immediately
- **Confidence Gate**: Technique confidence <50% **OR** remaining time/problems ratio ≥ 1.5 min/problem → SKIP
- **Strategic Skip**: Problems 16–20 with <10 solved → MARK for end
- **Time Pressure**: With 30 min left, attempt only problems with confidence >70%

120s Absolute Bailout:

- Mark: R (skip), technique tried, time spent
- Move immediately → no looking back
- Return only after attempting all accessible problems

AMC vs COMC Distinction:

- **AMC**: Can use constraint elimination for strategic letter guessing in bailout
- **COMC**: Bailout = leave blank and return later; no letter guessing → focus on objective bounds/diagram sanity only

2 Technique Recognition Cues

Section Overview and Usage

This section is a quick index organized into three tiers of cues: *Instant (1–2s)*, *Quick (4–10s)*, and *Fallback (pivot after 10–15s)*. Only the *Cue* → *Technique* → *First Move* box in Instant Recognition is WHO-only; the other tables include WHO/WHEN cues, with key signals folded into WHEN conditions. Each cue maps to a high-probability technique and a concrete first move. Use it as a speed aid; always confirm WHEN constraints/objectives before executing.

How to Use These Cues:

- Spot cue → confirm WHERE/WHEN → execute first move
- **During Practice**: Memorize 10–15 *Instant* cues; add 5–8 *Quick* cues for coverage
- **During Competition**: Try *Instant* first (1–2s); if none, scan *Quick* (to 10s); otherwise pivot to *Fallback*

Abbreviation Legend:

- **SFFT**: Simon's Favorite Factoring Trick (T5)
- **PoP**: Power of a Point (T23)
- **ABT**: Angle Bisector Theorem (T10)
- **FLT**: Fermat's Little Theorem (T20)
- **CRT**: Chinese Remainder Theorem (T38)
- **AM-GM**: Arithmetic Mean-Geometric Mean Inequality (T7)

2.1 Instant Recognition Cues (2s)

1–2 s recognition

2.1.1 WHO-only cues

Cue → Technique → First Move

Cue (WHO)	Technique	First Move
$xy + ax + by(+c)$	T5-SFFT	Add/subtract constant; factor to $(x+\alpha)(y+\beta)$
$x + \frac{c}{x}$ with $x > 0$	T7-AMGM	Bound; try equality at $x = \sqrt{c}$
Circle/tangent/secant/chords	T23-PoP	Apply tangent–secant/chord product
Angle bisector/side partition	T10-ABT	Set $\frac{BD}{DC} = \frac{AB}{AC}$
Digits/last k digits/divisibility	T19-ModArith	Work mod base/powers
Expected value / expected count	T11-LOE	Sum indicators; avoid independence
Overlap/union ("at least one")	T21-PIE	Tabulate intersections; add/subtract
Identical balls into boxes	TM1-StarsBars	Solve nonnegative integer equation
Remainder on polynomial division	T30-PolyDiv	Evaluate at root; match coefficients
Symmetric sums ($x+y,\ xy$ known)	T16-Vieta	Form quadratic; solve then back-substitute
Cyclic quad / tangent–chord	T23-PoP	Use equal angles; apply power relation
Telescoping-looking series	T29-Telescoping	Partial fractions/differences; sum
Polygon area from coordinates	TM3-Shoelace	Order vertices; apply shoelace sum; take $\frac{1}{2}$ absolute value
Cyclic quadrilateral side/diagonal relation	TM7-Ptolemy	Verify cyclic; apply $AC \cdot BD = AB \cdot CD + AD \cdot BC$
Two tangents from one point to a circle	TM11-CircleTangency	Mark radii; use $PA = PB$; replace tangent–chord with inscribed angle
$a^4 + 4b^4$ (or disguised)	TM18-SGI	Factor as $(a^2 - 2ab + 2b^2)(a^2 + 2ab + 2b^2)$
$(p-1)!$ (mod p) with p prime	TM19-Wilson	Pair inverses; conclude $(p-1)! \equiv -1 \pmod{p}$
Pipes/faucets/combined work or average speeds	TM23-RatesWork	Sum rates; use $d = rt$; harmonic mean for equal-distance round trip

Use as speed index: spot cue → confirm WHERE/WHEN (constraints/objective) → execute or pivot.

2.1.2 WHO/WHEN cues

Primary Cue(s) (WHO/WHEN)	**Technique (No. + Name)**	**First Move (link to setup)**
Repeated chunk; expression appears in multiple places; nested forms $f(g(x))$	T1-AlgSub	Set $u =$ (repeated chunk); rewrite equations/inequalities; normalize signs; see setup
Quadratic forms; circles/parabolas; sum of squares + linear terms	T8-CompleteSquare	Factor leading coeff; complete square; read vertex/center; see setup
Last digits; base-b representation; repeating decimals; digital sums	T9-BaseDigits	Work in base b and $b-1$; convert forms; track carries; see setup
Linear recurrences (order 1–2); constant coefficients	T13-CharEq	Write r-equation; solve; fit constants to initial values; see setup
Powers mod prime p; $a^{p-1} \equiv 1 \pmod{p}$	T20-FLT	Reduce exponent mod $p-1$; handle $a \equiv 0$ separately; see setup
Sums of squares/reciprocals; dot-product form; two sequences	T22-CauchySchwarz	Match vectors; apply C-S or Titu; simplify equality case; see setup
$\lfloor\cdot\rfloor, \lceil\cdot\rceil$ present; bounds; modular ties	T32-FloorCeil	Use inequalities $x - 1 < \lfloor x \rfloor \leq x$; split by thresholds; see setup
Integer-coefficient polynomial; small integer/rational roots likely	T33-RationalRoot	Test divisors of constant/leading; factor; see setup
Three cevians concurrent	T36-Ceva	Set product of ratios = 1; solve for unknown ratio; see setup
Three collinear points across triangle sides	T37-Menelaus	Apply ratio product $= -1$ (signed); solve for segment; see setup
Divide by linear factor; remainder quickly	T40-SyntheticDiv	Perform synthetic grid; read quotient/remainder; see setup
Colorings up to rotation/reflection; necklaces/bracelets	TM2-BurnsidePolya	List group elements; count fixed under rotations/reflections; average; see setup
Lattice polygon (integer-coordinate vertices); area/count	TM4-Picks	Compute B via edge gcds; use $A = I + \frac{B}{2} - 1$; see setup
Two denominations (packs/coins); representability/maximum gap	TM5-Frobenius	Check gcd; reduce; largest non-representable $ab - a - b$ (coprime); congruence filter; see setup
Modular inverse; linear congruence $ax \equiv b \pmod{m}$	TM6-ExtEuclid	Run EEA; invert if $\gcd(a, m) = 1$; solve and parametrize solutions; see setup
Incidence counts (pairs/endpoints/subsets); symmetric two-way counts	TM8-DoubleCount	Define incidence set; count two ways; equate (handshake lemma); see setup

2.2 Quick Recognition Cues (6s)

4–10 s recognition — *For setup steps see Technique Selection + Setup (6s)*

Primary Cue(s) (WHO/WHEN)	**Technique (No. + Name)**	**First Move (link to setup)**
Messy ratios/lengths; midpoints/slopes; circles; symmetry amenable to axes	T2-CoordGeo	Place coordinates at symmetry; define points; write equations; solve system; see setup
Small discrete ranges; parity/ sign/ ordering constraints; piecewise conditions	T3-CaseAnalysis	Split into minimal exhaustive cases; tabulate; prune impossible; see setup
Even/odd structure; symmetric sums; regular/palindromic digits or figures	T4-Symmetry	Reduce to fundamental region/variable; mirror/rotate; equalize pairs; see setup
More objects than boxes; guarantee of repetition/overlap	T6-Pigeonhole	Identify objects/boxes; compare counts; conclude existence/lower bound; see setup
Hard-to-count set equals an easier set; natural pairing	T12-Bijection	Construct 1-1 map; count target set instead; see setup
Count coprimes / reduce exponents modulo $\varphi(n)$; multiplicativity	T14-EulerPhi	Factor n; compute $\varphi(n)$; use exponent cycles; see setup
Max/min of sums of pairings; sorted sequences	T15-RearrangeInequ	Sort same/opposite order; bound sum; check extremal pairing; see setup
Random points/lengths/areas/angles; ratio-of-measures	T17-GeoProb	Express probability as measure ratio; integrate if needed; see setup
Rotations/dilations; oriented angles; circle geometry	T18-ComplexNumber	Encode points as complex; use arguments/moduli; translate to algebra; see setup
Expressions with $\sin, \cos, \tan$ or Pythagorean forms; angles	T24-TrigSub	Use identities (Pythagorean, sum-product); substitute $t = \tan\frac{\theta}{2}$ if algebraic; see setup
Sums $S_k = \sum x_i^k$ with relations; Newton sums	T27-PowerSum	Write Newton identities; relate coefficients and power sums; see setup
Permutations with no fixed points; "none in own spot"	T28-Derangements	Use $!n = n!\sum(-1)^k/k!$ or recursion; approximate with $\frac{n!}{e}$; see setup
Concurrent cevians; segment ratios along sides	T35-MassPoints	Assign masses; compute intersections; recover ratios/lengths; see setup

Multiple congruences; coprime moduli	T38-CRT	Solve pairwise; construct with inverses; recombine to modulus product; see setup
Adjacency/conflict/pairing constraints; networks	T39-GraphModel	Model with vertices/edges; degrees/handshake; components; see setup
Two sides + included angle; three sides; need R or area via sin	TM9-TriTrig	Apply LoC/LoS; compute $R = \frac{a}{2\sin A}$; use $A = \frac{1}{2}ab\sin C$; see setup
Side lengths/semiperimeter; incircle/circumcircle; cyclic quad area	TM10-AreaToolkit	Compute s; use Heron/A=rs/abc/(4R); if cyclic quad, Brahmagupta; see setup
Parameter bounds from real roots/tangency/nonnegativity	TM12-Discriminant	Write Δ; set $\Delta \geq 0$ or ≤ 0; solve for parameters; see setup
Products/ratios, exponent/log equations, telescoping logs	TM13-Logs	Normalize base; expand/compress with log rules; ensure domain; see setup
"At least one"/complements easier than direct	TM14-Complement	Count complement; apply PIE if overlaps; subtract from total; see setup
Faces/edges/vertices of convex polyhedra or planar graphs	TM15-EulerPoly	Use $V-E+F = 2$; combine with $\sum \deg(v) = 2E$; see setup
Three tangent circles plus one unknown (or an enclosing circle)	TM16-Descartes	Assign curvatures with correct signs; solve quadratic for unknown; see setup
Real-coefficient polynomial with non-real roots	TM17-ConjRoots	Pair $re^{\pm i\theta}$ to form real quadratic; factor/deflate; see setup
$v_p(n!)$ or $v_p\binom{n}{k}$; factorial/binomial exponents	TM20-Legendre	Sum floors; subtract for binomials; stop when $p^k > n$; see setup
Number of divisors or sum of divisors of n	TM21-DivFuncs	Factor n; apply τ and σ prime-power formulas; see setup
Coefficient extraction / binomial sums / layered selections	TM22-BinomialPascal	Choose binomial/Pascal/Vandermonde; reindex carefully; see setup

2.3 Fallback Recognition Cues

Pivot after 10–15 s — *Use when no instant/quick cue fits; higher-setup conversions*

Primary Cue(s) (WHO/WHEN)	**Technique (No. + Name)**	**First Move (link to setup)**

Composition/counting with order or unlimited copies	T25-GenFunc	Encode as product; extract coefficient of x^n; partial fractions if simple; see setup
Process of moves; parity/coloring; never-decreasing quantity	T26-Invariants	Identify invariant/monovariant; prove impossibility or termination; see setup
Triangle with cevian; split BC into $BD = m,\ DC = n$ $(m{+}n = a)$	T31-Stewart	Use $b^2m + c^2n = a\big(d^2 + mn\big)$; see setup
Convex/concave f with fixed sum/mean	T34-JensenInequ	Verify convexity; apply at mean; equality at equal variables; see setup
$f(x+y)$, $f(ax)$, $f(f(x))$; injective/surjective hints	T41-FuncEq	Plug $0, 1, -x$; compare $f(a) - f(b)$; search fixed points; see setup

3 Strategic Time Management

Integration with Rapid Classification System

This section is fully aligned with the Rapid Problem Classification System (Section 1)
Universal Timeline for Every Problem:

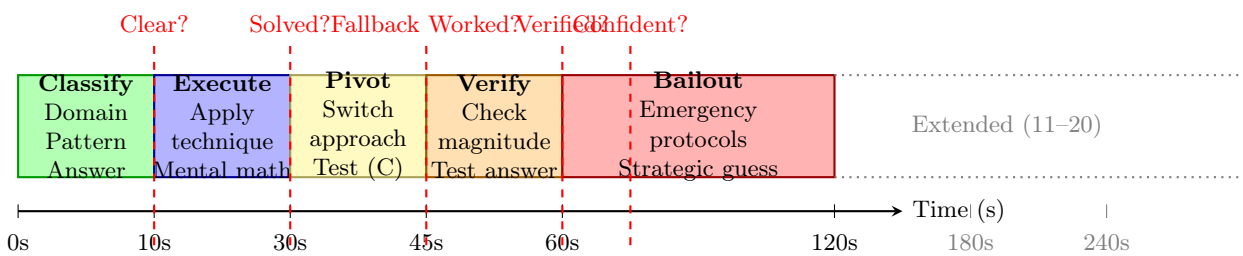
- **0–15s**: Classification (15-Second Method)
- **15–35s**: Execution (Domain Lanes)
- **35–50s**: Pivot if needed (35-Second Pivot)
- **50–65s**: Verification (Answer Testing)
- **65–120s**: Emergency protocols (Red Flags)

Key Principle: The same classification timeline applies to ALL problems (1–20), but total time budgets scale with difficulty:

- Problems 1–10: Complete most within 65–120s
- Problems 11–15: Allow 180–240s with extended execution
- Problems 16–20: May need 240–300s for complex work

Navigation: Each time management subsection below includes hyperlinks back to the relevant classification phases.

3.1 Visual Timeline Overview



Problems 1–10: Usually complete by 65s
Problems 11–15: May extend to 180s
Problems 16–20: May extend to 240s

3.2 Time Budget Guidelines with Pattern Recognition

Time Allocation by Problem Range - Aligned with Rapid Classification

Problem #	Target Time	Classification	Success Rate	Strategy
1–5	65–120s	0–15s	90–95%	High-Prob Conv.
6–10	120–180s	0–15s	85–92%	High-Prob Conv.
11–15	180–240s	0–15s	75–85%	Tiered Conv.
16–20	240–300s	0–15s	60–75%	Mixed Strategy

Alignment with Rapid Classification Timeline:

- **0–15s**: Classification phase (domain scan + pattern match + answer intelligence)
- **15–35s**: Domain-specific execution lanes
- **35–50s**: Pivot strategy if stuck
- **50–65s**: Answer testing and verification
- **65–120s**: Emergency protocols and bailout

Notes: Success rates are indicative for well-prepared students. The **Classification** time includes the complete 10-second classification method from Section 1.

Detailed Time Breakdown - Integrated with Classification System

Phase	Classify (0–15s)	Execute (15–35s)	Pivot (35–50s)	Verify (50–65s)	Bailout (65–120s)
Very Early (1–5)	15s	20s	15s	15s	30s
Early (6–10)	15s	20s	15s	15s	40s
Mid (11–15)	15s	20s	15s	15s	45s
Late (16–20)	15s	20s	15s	15s	60s

Note: This table reflects the initial 0–120s classification cycle. Mid/late-band problems may extend beyond this window; see the extended work phases described below.

Phase Descriptions (aligned with Section Timeline Legend):

- **Classify**: Domain scan + pattern match + answer intelligence (see 15-Second Classification)
- **Execute**: Apply domain-specific technique (see Domain Lanes)
- **Pivot**: Switch technique if stuck (see 35-Second Pivot)
- **Verify**: Test answers and validate (see 50-Second Testing)
- **Bailout**: Emergency protocols (see Red Flags)

Per-Problem Hard Stops - Decision Gate Framework

- **Problems 1–10**: Maximum 2 minutes per problem
  - **15s**: Classification complete → proceed or skip
  - **35s**: Execution check → continue or pivot
  - **65s**: Answer selected or mark for return
  - **120s**: Absolute maximum → move on
- **Problems 11–15**: Maximum 3.5 minutes per problem
  - Follow same gates but allow extra time in execution phase
  - **180s**: Begin wrap-up regardless of progress
- **Problems 16–20**: Maximum 4.5 minutes per problem
  - May extend execution to **150s** if making clear progress
  - **240s**: Must enter verification/bailout phase

Decision Gate Criteria (from Timeline Legend):

- **Clear?** Domain and technique identified with confidence
  - YES → Move to Domain Lanes (Identify domains and techniques for execution)
  - NO → Move to Pivot Strategy (need different approach)
- **Solved?** Made meaningful progress with domain-specific technique
  - YES → Move to Answer Verification (solution ready for testing)
  - NO → Move to Pivot Strategy (need different approach)
- **Fallback Worked?** Pivot technique providing clear progress and manageable calculation
  - YES → Move to Answer Verification (made progress, ready to test solution)
  - NO → Move to Answer Elimination (systematic rule-out as backup approach)
- **Verified?** Answer verification successful during testing phase
  - YES → Proceed directly to Next Problem (skip elimination)
  - NO → Move to Answer Elimination for systematic rule-out
- **Confident?** Answer appears solid and constraints are satisfied
  - YES → Proceed to Next Problem (answer is reliable)
  - NO → Enter Red Flags protocols for deeper verification

Run-Wide Pacing and Buffers (75 minutes)

Overall Competition Timeline:

- **Minutes 0–22**: Problems 1–10 using Rapid Classification (Section 1.1)
- **Minutes 22–40**: Problems 11–15 with extended execution phases

- **Minutes 40–60**: Problems 16–20 or strategic revisits
- **Minutes 60–72**: High-value returns and verification
- **Minutes 72–75**: Bubble sheet and final scan

Progress Checkpoints:

- **20 min**: Should have attempted 8–10 problems
- **40 min**: Should have attempted 13–15 problems
- **60 min**: Should have attempted 17–20 problems
- If behind schedule, prioritize problems matching your strengths

Strategic Time Reserve:

- Keep 10–15 minutes for selective revisits
- Return to YELLOW-marked problems first (from classification)
- Use 50-Second Answer Testing for quick gains
- Never leave answer sheet blank if you've eliminated any choices

3.3 Early-Band Time Management (Problems 1–10)

Problems 1–10: Rapid Classification Timeline Integration

Time Allocation Strategy (65–120s per problem):

1. **[0–15s] Classification Phase** (15-Second Method)
   - **0–4s**: Domain scan → tag ALG/GEO/CNT/NT/PROB
   - **4–10s**: Pattern match using WHO/WHEN framework
   - **10–15s**: Answer intelligence → GREEN/YELLOW/RED decision
2. **[15–35s] Execution Phase** (Domain Lanes)
   - Apply selected technique immediately
   - Use mental math whenever possible
   - Monitor progress at 15s, 20s, 25s checkpoints
3. **[35–50s] Pivot if Needed** (35-Second Pivot)
   - Only enter if stuck at 35s checkpoint
   - Try simpler approach or apply constraints to eliminate choices
4. **[50–65s] Verification** (50-Second Testing)
   - Quick magnitude check if confident
   - Apply constraint elimination if uncertain

Early-Band Efficiency Rules:

- **Trust the classification**: If GREEN at 15s, execute immediately
- **Mental math priority**: Write only when necessary
- **Skip heavy verification**: If confidence $> 90\%$, move on
- **65s hard limit**: Never exceed 65 seconds for problems 1–5
- **120s absolute max**: Problems 6–10 get extra time only if near solution

Common Time Wasters to Avoid:

- Over-reading after classification complete
- Trying multiple methods instead of committing
- Excessive algebraic manipulation
- Full verification when pattern is clear

Example: Early-Band Problem Timeline

Problem: Find the value of $\frac{2^{10} \cdot 5^8}{10^7}$.
Timeline Execution:

- **[0–4s]**: Domain scan → ALG (exponent manipulation)
- **[4–10s]**: Pattern match → "Simplify using common base"
- **[10–15s]**: Answer scan → Powers of 2 likely → GREEN
- **[15–25s]**: Execute: $\frac{2^{10} \cdot 5^8}{10^7} = \frac{2^{10} \cdot 5^8}{(2 \cdot 5)^7} = \frac{2^{10} \cdot 5^8}{2^7 \cdot 5^7} = 2^3 \cdot 5 = 40$
- **[20–25s]**: Quick verify: $8 \cdot 5 = 40$ ✓
- **Total**: 25 seconds, move to next problem

Key Efficiency: Mental math only, no written work needed. Pattern recognized immediately led to correct technique.

Transition Efficiency Between Problems:

- **Physical transition** (2–3s): Mark answer, flip page, reset mental state
- **Mental reset**: Clear previous problem completely before reading next
- **Answer sheet strategy**: Mark after every 3–4 problems to batch interruptions
- **Workspace management**: Use consistent areas of scratch paper for each problem
- **Energy management**: Take one deep breath between problems to maintain focus

3.4 Mid-Band Time Management (Problems 11–20)

Problems 11–20: Extended Classification Timeline

Time Allocation Strategy (180–300s per problem):

1. **[0–15s] Classification Phase** - Same as early-band but more critical
   - Domain scan with extra care for mixed-domain problems
   - Pattern recognition may identify multiple viable techniques
   - Answer analysis crucial for technique selection
2. **[15–35s] Initial Execution**
   - Set up problem with clear notation
   - Apply primary technique systematically
   - Document key steps for potential backtracking
3. **[35–50s] Assessment and Pivot**
   - **Progress check**: Are you advancing toward solution?
   - If stuck: Switch to secondary technique from classification
   - If complex: Consider coordinate geometry or casework
4. **[50–65s] Intensive Verification**
   - Test answer in original problem
   - Check boundary cases
   - Verify all constraints satisfied
5. **[60–180s] Extended Work Phase** (Problems 11–15)
   - **When to use**: Enter this phase only if making clear progress at 65s

- Continue systematic execution with current technique
- May attempt one alternative technique if primary fails
- Use answer choices to guide/verify work
- **Exit criteria**: Solution found OR 180s reached OR no further progress

6. **[60–240s] Complex Problem Phase** (Problems 16–20)
   - **When to use**: Reserve for problems where partial progress is evident
   - Accept that full solution may be needed
   - Build partial progress toward elimination
   - Strategic guessing if time pressure mounts
   - **Exit criteria**: Solution found OR 240s reached OR can eliminate 2+ choices

Mid-Band Decision Framework:
- **At 35s**: Solved? → Continue to verification, NO → Pivot strategy
- **At 65s**: Verified? → Next problem, NO → Answer elimination
- **At 120s**: For 11–15, should be near solution or testing answers
- **At 180s**: For 16–20, make educated guess if not converging

Efficiency Techniques:
- **Answer elimination first**: Narrow to 2–3 choices before heavy computation
- **Estimation**: Get magnitude, then refine
- **Work backwards**: Apply constraints to narrow choices, then test promising candidate
- **Partial solving**: Distinguish between remaining choices

Skill Level Adjustments:
- **Well-prepared**: Use times as shown
- **Intermediate**: Add 30–60s per problem
- **Beginner**: Focus on problems 11–15, skip 16–20 initially
- **Advanced**: Subtract 15–30s, attempt all problems

Example: Mid-Band Problem Timeline

Problem: How many ordered pairs of positive integers (a, b) satisfy $\frac{1}{a} + \frac{1}{b} = \frac{1}{12}$?

Timeline Execution:

- **[0–4s]**: Domain scan → NT/CNT (Diophantine equation)
- **[4–10s]**: Pattern match → "Reciprocal equation, factor manipulation"
- **[10–15s]**: Answer scan → Small count expected → YELLOW (needs algebra)
- **[15–35s]**: Initial execution:
  - Multiply through: $12b + 12a = ab$
  - Rearrange: $ab - 12a - 12b = 0$
  - Add 144: $ab - 12a - 12b + 144 = 144$
- **[35–50s]**: Recognition and pivot:
  - Factor: $(a - 12)(b - 12) = 144$
  - Need positive a, b so $a > 12, b > 12$
- **[45–75s]**: Complete solution:
  - Divisors of 144: 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144
  - Count positive divisor pairs: 15 pairs
  - Each gives valid (a, b) with $a = d_1 + 12, b = d_2 + 12$
- **[75–90s]**: Verify with one case: $d_1 = 1, d_2 = 144 \to a = 13, b = 156$
  - Check: $\frac{1}{13} + \frac{1}{156} = \frac{12}{156} + \frac{1}{156} = \frac{13}{156} = \frac{1}{12}$ ✓
- **Total**: 90 seconds, answer is 15

Key Decision Points:

- At 35s: Recognized Simon's Favorite Factoring Trick pattern
- At 50s: Committed to counting divisors rather than trying cases
- Verification kept brief—one test case sufficient

3.5 Time Pressure Decision Points

Critical Time Checkpoints - Aligned with Classification Gates

Timeline	Gate	Early-Band (1–10)	Mid-Band (11–20)
15s	Clear?	Classification complete Proceed if GREEN	Classification complete Consider multiple techniques
35s	Solved?	Execution checkpoint YES → Verify, NO → Pivot	Initial execution done Assess progress critically
50s	Fallback Worked?	Begin verification YES → Verify, NO → Eliminate	Pivot or intensive verify Try secondary technique
65s	Verified?	Final answer or skip YES → Next problem, NO → Eliminate	Enter extended phase Continue if progressing
65s	Confident?	Final answer or skip YES → Next problem, NO → Red flags	Enter extended phase Continue if progressing
120s	Bailout	Absolute maximum Mark and skip	Mid-band checkpoint Should be near solution

Decision Gate Actions:

- **Clear? (15s)**: Domain identified, technique selected
  - YES → Execute domain-specific technique
  - NO → Move to Pivot Strategy
- **Solved? (35s)**: Made meaningful progress with domain-specific technique
  - YES → Move to Answer Verification
  - NO → Move to Pivot Strategy
- **Fallback Worked? (50s)**: Pivot technique providing clear progress and manageable calculation
  - YES → Move to Answer Verification
  - NO → Move to Answer Elimination
- **Verified? (65s)**: Answer verification successful during testing phase
  - YES → Proceed directly to Next Problem
  - NO → Move to Answer Elimination
- **Confident? (65s)**: Answer appears solid and constraints are satisfied
  - YES → Proceed to Next Problem
  - NO → Enter Red Flags protocols
- **Bailout (120s)**: Time limit reached → mark and move on

Strategic Guessing Thresholds:

- **Eliminated 0 choices**: Skip (blank = 1.5 pts)
- **Eliminated 1 choice**: Break-even (guess from 4 = 1.5 pts)
- **Eliminated 2 choices**: Guess (EV from 3 = 2.0 pts)
- **Eliminated 3+ choices**: Always guess (EV $\geq$ 2.5 pts)

AMC12 Expected Value: Blank = 1.5, Random from 5 = 1.2, Strategic elimination improves EV significantly.

3.6 Problem Order Strategy

Strategic Problem Navigation

Default Strategy: Modified Sequential

- **Problems 1–10**: Work sequentially, skip only if RED classification
- **Problems 11–15**: Quick scan all 5, start with strongest domain
- **Problems 16–20**: Cherry-pick based on domain strengths
- **Final 15 minutes**: Return to YELLOW-marked problems

When to Skip vs. Persist:

- **Skip immediately** (within 10s):
  - RED classification with no clear approach
  - Domain you consistently struggle with
  - Problem requires technique you haven't mastered
- **Persist** (up to time limit):
  - GREEN classification with clear technique
  - Making measurable progress each 15s
  - Can eliminate 2+ answer choices
- **Mark and return** (YELLOW):
  - Technique clear but calculation lengthy
  - Need to verify a specific case
  - One insight away from solution

Strategic Jumping Patterns:

1. **Geometry Specialist**: After problem 10, scan 11–20 for GEO tags
2. **Algebra/Calculus Strong**: Prioritize ALG problems in 11–15 first
3. **Counting Expert**: Jump to CNT/PROB problems regardless of position
4. **Number Theory Weak**: Save NT problems for end or skip entirely

Return Priority Matrix:

Problem State	Time Investment	Return Priority
Eliminated 3+ choices	30–60s	HIGH
Calculation incomplete	60–90s	MEDIUM
Need to verify	50–65s	MEDIUM
Stuck at setup	30s+	LOW
No progress	Any	SKIP

3.7 Time Recovery Protocols

Behind Schedule Recovery

If Behind at 20-Minute Mark (fewer than 8 problems attempted):

1. Immediately skip to problems 6–10 if still in 1–5
2. Reduce verification phase to 5s magnitude checks only
3. Use answer (C) testing more aggressively
4. Accept 80% confidence instead of 95%

If Behind at 40-Minute Mark (fewer than 13 problems attempted):

1. Abandon problems 16–20 entirely
2. Focus on completing 11–15 with your strongest domains
3. Switch to "elimination mode": get to 2–3 choices and guess
4. Reserve last 20 minutes for high-probability returns only

If Behind at 60-Minute Mark (fewer than 17 problems attempted):

1. Stop attempting new problems
2. Return only to problems where you eliminated 2+ choices
3. Use pattern recognition for educated guesses
4. Ensure no blanks where you eliminated any choice

Emergency Speed Mode:

- 30s per problem maximum for remaining attempts
- Test answer (C) first, then (D) or (B) based on magnitude
- Use dimension analysis and magnitude estimation only
- Trust first instinct on pattern recognition

3.8 Score-Based Time Management

Adjusting Strategy by Target Score

Note: AMC12 scoring: 6 points per correct answer, 1.5 for blank, 0 for incorrect. Target scores assume high accuracy on attempted problems.

Target Score	Problems Focus	Time Distribution	Strategy
90–100	1–12 solid 2–3 from 13–15	70% on 1–10 30% on 11–15	Accuracy over speed Skip 16–20 entirely
100–110	1–13 solid 3–4 from 14–17	50% on 1–10 40% on 11–15 10% on 16–20	Balanced approach Sample 16–20
110–120	1–15 solid 2–3 from 16–20	40% on 1–10 35% on 11–15 25% on 16–20	Speed through early Full attempt 11–15 Strategic 16–20
120+	1–17 solid 2–3 from 18–20	35% on 1–10 35% on 11–15 30% on 16–20	Rapid early completion Efficient mid-band Serious attempt 16–20

Score-Specific Tips:

- **90–100 points**: Need 15–16 correct + blanks; accuracy on 1–10 is crucial
- **100–110 points**: Need 16–18 correct; don't sacrifice 1–10 accuracy for 16–20 attempts
- **110–120 points**: Need 18–19 correct; must maintain 90%+ accuracy on 1–10 while attempting most of 11–15
- **120+ points**: Need 19–20 correct; requires both speed and accuracy; rapid classification essential

4 Common Trap Patterns and Avoidance

Section Overview: Trap Pattern Recognition

This section identifies the most common trap patterns in AMC12/COMC problems and provides systematic avoidance strategies. Each trap includes:

- **Severity Rating**: High (catches 80%+), Medium (40-80%), Low (<40%)
- **Frequency**: How often the trap appears in competition problems
- **Recognition**: Key indicators that a trap may be present
- **Avoidance**: Specific strategies to prevent falling into the trap

Integration with Rapid Classification: Trap detection occurs primarily at:

- **10s**: Initial answer choice analysis for trap indicators
- **45s**: Verification phase to catch computational traps
- **65s**: Red flag detection for conceptual traps

Trap Icons Legend: △ = Computational trap, ★ = Conceptual trap, ⋄ = Geometric trap, □ = Algebraic trap

4.1 Trap Patterns in Early-Band Problems (1–10)

The "Obvious Answer" Trap High Severity △

Pattern: Answer that comes from misreading or incomplete calculation
Frequency: Appears in 25% of problems 1–5
Example: "Find 30% of 250" → Trap: 7.5 (decimal error) or 30 (just the percentage)
Correct answer: 75
Recognition: Answer choices include intermediate calculations or unit-less values
Avoidance: Always complete the full calculation; check units and magnitude; re-read what's being asked

The "Forgot to Finish" Trap High Severity ★

Pattern: Intermediate result is an answer choice
Frequency: Appears in 30% of geometry problems 1–10
Example: "Find the area of a square with diagonal 10" → Trap: 100 (that's diagonal^2)
Correct answer: 50 (Area = diagonal^2/2)
Recognition: Answer choices include values from intermediate steps
Avoidance: Highlight what's being asked; verify your answer addresses the actual question

The "Sign Error" Trap Medium Severity □

Pattern: Negative of correct answer is present
Frequency: Appears in 20% of algebra problems
Example: Solving $2x + 5 = -3$ where both $x = 4$ and $x = -4$ are choices
Recognition: Answer choices include positive/negative pairs
Avoidance: Track signs at each step; circle negative signs; verify by substitution

The "Off-by-One" Trap Medium Severity △

Pattern: Counting problems with fence-post errors
Frequency: Appears in 35% of counting problems
Example: "Numbers from 1 to 100 inclusive" → Trap: 99 instead of 100

Recognition: Answer choices differ by exactly 1
Avoidance: Use formula: inclusive count = last - first + 1; test with small cases

The "Wrong Operation" Trap Medium Severity △

Pattern: Used addition instead of multiplication (or vice versa)
Frequency: Appears in 20% of probability/counting problems
Example: Probability problems → Trap: Adding instead of multiplying
Recognition: Keywords "and" vs "or", independent vs dependent events
Avoidance: For "and", multiply only if events are independent; otherwise use conditional probability. For "or", add only if events are mutually exclusive; otherwise use inclusion–exclusion. Clarify whether order matters and whether sampling is with or without replacement.

The "Units and Conversions" Trap High Severity ◇

Pattern: Mixed or implicit units lead to wrong magnitude
Frequency: Appears in 25% of rate/measurement problems
Example: Traveling 50 mph for 30 minutes → Trap: 50
Correct answer: 25 miles
Recognition: Mixed time units, metric vs imperial, area/volume scaling
Avoidance: Convert units first (minutes to hours, degrees to radians, linear vs area vs volume units); label every number with its unit

The "Successive Percentage" Trap Medium Severity □

Pattern: Treating successive percent changes additively
Frequency: Appears in 15% of percent change problems
Example: $+20\%$ then -20% → Trap: net 0%
Correct answer: Net -4%
Recognition: Multiple percent changes in sequence
Avoidance: Multiply by factors: $(1+0.20)(1-0.20) = 0.96$; never add/subtract percentages directly

The "Average Speed" Trap Medium Severity ◇

Pattern: Averaging speeds instead of using time-weighted or harmonic mean
Frequency: Appears in 20% of rate problems
Example: Equal distances at 30 mph and 60 mph → Trap: 45 mph
Correct answer: 40 mph (harmonic mean for equal distances)
Recognition: Multiple segments with different speeds
Avoidance: Use total distance divided by total time; for equal distances: $\frac{2v_1v_2}{v_1+v_2}$; for equal times: simple average

The "Vieta Without Monic" Trap Medium Severity □

Pattern: Applying Vieta's formulas to non-monic polynomials without adjustment
Frequency: Appears in 20% of polynomial problems
Example: For $2x^2 - 6x + 4 = 0$, sum of roots → Trap: 6
Correct answer: 3 (divide by leading coefficient: $-(-6)/2 = 3$)
Recognition: Non-monic polynomial with root sum/product questions
Avoidance: Always make polynomial monic first or adjust formulas; for $ax^2 + bx + c$: sum $= -b/a$,

product $= c/a$
Reference: See Technique 16 (Vieta's Formulas) in Conversion Techniques guide

The "AM-GM Sign" Trap Medium Severity □

Pattern: Applying AM-GM to expressions with negative terms
Frequency: Appears in 20% of inequality/optimization problems
Example: Minimizing $x - \frac{1}{x}$ for $x > 0$ using AM-GM
Correct approach: AM-GM requires all terms non-negative
Recognition: Optimization with mixed signs; inequality application
Avoidance: Check positivity before applying; consider substitution if needed; use calculus or other methods for mixed-sign expressions
Reference: See Technique 7 (AM-GM Inequality) in Conversion Techniques guide

4.2 Trap Patterns in Mid-Band Problems (11–20)

The "Partial Solution" Trap High Severity ★

Pattern: Solving for x when problem asks for $x^2 + 1$
Frequency: Appears in 30% of mid-band algebra problems
Example: Quadratic gives $x = 3$, but question asks for $x^2 + 2x$
Recognition: Answer choices include both x values and expressions in x
Avoidance: Highlight what's being asked; write "FIND: [expression]" at top of work; check before submitting

The "Special Case" Trap Medium Severity △

Pattern: General answer differs from specific cases
Frequency: Appears in 25% of sequence/induction problems
Example: "For all positive integers n..." but $n = 1$ is special
Recognition: Problem states "for all n" or "$n \geq 1$"
Avoidance: Always test $n = 1, n = 2$ as verification; check if formula needs adjustment for small cases

The "Overcounting" Trap Medium Severity □

Pattern: Counting arrangements with hidden restrictions
Frequency: Appears in 30% of combinatorics problems
Example: Circular arrangements, identical objects, symmetry
Recognition: Keywords like "circular", "identical", "indistinguishable"
Avoidance: Circular: divide by n for rotations; identical: use multinomial coefficients; check for symmetry duplicates

The "Inclusion–Exclusion" Trap

Pattern: Double-counting unions of sets or outcomes
Example: $|A \cup B| \to$ Trap: $|A| + |B|$
Avoidance: Use $|A \cup B| = |A| + |B| - |A \cap B|$; extend to 3 sets if needed

The "Extraneous Solutions" Trap

Pattern: Solutions introduced by squaring, clearing denominators, or absolute values
Example: Solving $\sqrt{x} = x - 2$ yields $x = 1, 4$ after squaring $\rightarrow$ Trap: accept both
Avoidance: Record domain restrictions up front; check candidates in the original equation (only $x = 4$ works here)

The "Absolute Value Cases" Trap

Pattern: Treating $|\cdot|$ equations/inequalities as single-case
Example: $|x - 3| = 7 \rightarrow$ Trap: $x = 10$ only
Avoidance: Split into cases: $x - 3 = 7$ or $x - 3 = -7$; for inequalities, translate to intervals and check endpoints

The "Permutation vs Combination" Trap

Pattern: Ignoring whether order matters
Example: Choosing a 3-person committee vs assigning president, VP, secretary
Avoidance: Use $\binom{n}{k}$ for selections; use $P(n, k) = \frac{n!}{(n-k)!}$ when order matters; in circular arrangements, account for rotations (and reflections when relevant)

The "Scaling" Trap

Pattern: Confusing how lengths, areas, and volumes scale under similarity
Example: Side doubled $\rightarrow$ Trap: area doubles
Avoidance: Lengths scale by k; areas by k^2; volumes by k^3

The "Domain/Constraint" Trap

Pattern: Ignoring problem constraints or implicit domains
Example: Taking logs of nonpositive numbers; dividing by an expression that could be 0
Avoidance: State domain and constraints before solving; enforce them when selecting answers

The "Symmetry/Parity" Trap

Pattern: Missing invariants or symmetry reduces work and avoids double-counting
Example: Counting arrangements where swapping symmetric positions produces duplicates
Avoidance: Identify invariants, parity, or symmetry classes; reduce to canonical representatives

The "Power of a Point Configuration" Trap

Pattern: Misidentifying whether point is inside or outside circle
Example: Using tangent formula $PT^2 = PA \cdot PB$ for internal point
Correct approach: Internal: $PA \cdot PB = PC \cdot PD$; External: $PT^2 = PA \cdot PB$
Avoidance: Check point position relative to circle before applying formula
Reference: See Technique 23 (Power of a Point) in Conversion Techniques guide

The "Cauchy-Schwarz Equality" Trap

Pattern: Assuming maximum is always achievable under constraints
Example: Maximizing $3x + 4y$ with $x^2 + y^2 = 25$ and $x \geq 0$
Correct approach: Check if proportionality $\frac{x}{3} = \frac{y}{4}$ satisfies all constraints
Avoidance: Verify equality conditions are feasible within problem constraints

Reference: See Technique 22 (Cauchy-Schwarz) in Conversion Techniques guide

The "Linearity of Expectation Product" Trap

Pattern: Assuming $E[XY] = E[X] \cdot E[Y]$ without independence
Example: Expected value of product of dependent random variables
Correct approach: Linearity applies to sums, not products (unless independent)
Avoidance: Use indicators for counting; check independence for products
Reference: See Technique 11 (Linearity of Expectation) in Conversion Techniques guide

The "Bijection Without Proof" Trap

Pattern: Assuming a correspondence is bijective without verification
Example: Claiming two sets have equal size via unverified mapping
Correct approach: Prove both injectivity and surjectivity
Avoidance: Explicitly construct inverse map or show one-to-one and onto
Reference: See Technique 12 (Bijection Principle) in Conversion Techniques guide

4.3 Additional High-Priority Trap Patterns

The "Complex Conjugate" Trap Medium Severity □

Pattern: Forgetting complex roots come in conjugate pairs for real polynomials
Frequency: Appears in 15% of problems 15–20
Example: If $2 + 3i$ is a root of a real polynomial, missing that $2 - 3i$ must also be a root
Recognition: Complex number appears as a root; polynomial has real coefficients
Avoidance: For real polynomials, complex roots always come in conjugate pairs; factor as $(x - (a + bi))(x - (a - bi)) = x^2 - 2ax + (a^2 + b^2)$
Reference: See Technique 26 (Complex Numbers) in Conversion Techniques guide

The "Trigonometric Quadrant" Trap High Severity ◇

Pattern: Sign errors when working with trig functions in different quadrants
Frequency: Appears in 25% of trig problems
Example: Finding $\sin\theta$ given $\cos\theta = -\frac{3}{5}$ and θ in Q2 $\rightarrow$ Trap: $-\frac{4}{5}$
Correct answer: $+\frac{4}{5}$ (sine is positive in Q2)
Recognition: Problem involves angles outside Q1; multiple angle solutions
Avoidance: Use ASTC (All Students Take Calculus) for quadrant signs; sketch unit circle
Quick Reference: Q1: all positive, Q2: sin positive, Q3: tan positive, Q4: cos positive

The "Functional Equation Injectivity" Trap Medium Severity ★

Pattern: Assuming a function is injective/surjective without verification
Frequency: Appears in 20% of problems 16–20
Example: From $f(f(x)) = x$, concluding f is its own inverse without checking bijectivity
Recognition: Functional equations with composition; "find all functions" problems
Avoidance: Test specific values (0, 1, -1); check if f is both one-to-one and onto; consider piecewise functions as counterexamples
Reference: See Technique 27 (Functional Equations) in Conversion Techniques guide

The "Logarithm Base" Trap Medium Severity ☐

Pattern: Confusing natural log, common log, and arbitrary base logarithms
Frequency: Appears in 15% of exponential/log problems
Example: Using $\log_2 8 = 3$ but treating it as $\ln 8$ or $\log_{10} 8$
Recognition: Multiple logarithms with different or unspecified bases
Avoidance: Always identify base; use change of base formula: $\log_a b = \frac{\log_c b}{\log_c a}$; remember log alone usually means $\log_{10}$ in competitions

The "Modulus vs Argument" Trap Low Severity ☐

Pattern: Confusing magnitude and angle in complex numbers
Frequency: Appears in 10% of complex number problems
Example: For $z = 3e^{i\pi/4}$, using 3 as the argument instead of the modulus
Recognition: Polar form complex numbers; geometric interpretations
Avoidance: $|z| = r$ (distance from origin), $\arg(z) = \theta$ (angle from positive real axis); for $z = re^{i\theta}$, modulus is r, argument is θ

4.4 Universal Trap Avoidance Checklist

Enhanced Trap Detection Protocol with Time Estimates

Before Starting [3–5 seconds]:
- ☐ **[1s]** Read the ENTIRE problem (including what's asked)
- ☐ **[1s]** Note any keywords: "positive", "integer", "distinct", "at least", "exactly"
- ☐ **[1s]** Check answer choice magnitudes for reasonableness
- ☐ **[1s]** Record domain/constraints: nonnegative, integer, distinct, denominators $\neq 0$, radicands ≥ 0
- ☐ **[1s]** Underline/annotate units; convert early if helpful

During Solving [continuous monitoring]:
- ☐ **[2s]** Write down what you're solving for (at start)
- ☐ **[ongoing]** Keep units/labels with numbers
- ☐ **[ongoing]** Don't skip "obvious" algebraic steps
- ☐ **[2s]** For probability/counting: mark whether events are independent/disjoint; order and replacement
- ☐ **[1s]** When squaring/clearing denominators/handling $|\cdot|$, flag to check for extraneous solutions

Before Committing [5–10 seconds]:
- ☐ **[1s]** Re-read the question stem
- ☐ **[2s]** Verify answer satisfies all constraints
- ☐ **[1s]** Check if answer magnitude makes sense
- ☐ **[1s]** If your answer isn't there, you likely fell for a trap
- ☐ **[3s]** Substitute into the original equation/definition to verify (if time permits)
- ☐ **[2s]** Quick estimate or extreme-case check to sanity-check magnitude

Time Investment:
- Problems 1–10: Spend 3–5s on pre-checks, 5s on final verification
- Problems 11–15: Spend 5s on pre-checks, 10s on verification
- Problems 16–20: Full protocol if time permits (15–20s total)

4.5 Answer Choice Analysis for Trap Detection

Red Flags in Answer Choices

1. **Too many "nice" numbers**: One is probably a trap
2. **Arithmetic sequence**: Often includes partial results
3. **Powers of 2 or 10**: Check if you completed all steps
4. **Negative and positive versions**: Sign error trap likely
5. $\sqrt{n}$ **and** n: Check if you need to square/take root
6. **Reciprocal forms** x **vs** $1/x$: Check for inversion or harmonic mean traps
7. **Both "order" and "no-order" counts present**: Decide permutation vs combination
8. **One value off by 1**: Suspect fence-post or inclusive/exclusive endpoint errors
9. **Both linear and squared forms**: You may have stopped at an intermediate (unsquared) step
10. **Mixed units or scaled answers**: Confirm unit conversions and similarity scaling
11. **An "integerized" value**: If problem constraints require integer, ensure you enforced them; otherwise beware rounding traps

Strategic Use:

- If stuck, identify likely traps first
- Eliminate trap answers to improve guessing odds
- Work backwards from non-trap answers

4.6 Trap Recovery Strategies

When You Realize You've Fallen Into a Trap

Recognition Signs [2–3 seconds to identify]:

- Your answer matches a "too obvious" choice
- Calculation becomes unexpectedly complex
- Answer violates problem constraints
- Time spent exceeds expected for problem difficulty

Recovery Protocol by Time Remaining:

If >65s remaining on problem:

1. **[5s]** Mark current work with "TRAP?"
2. **[10s]** Re-read problem statement completely
3. **[15s]** Identify trap type from answer patterns
4. **[20s]** Apply specific counter-strategy
5. **[10s]** Verify new approach addresses original question

If 35–65s remaining:

1. **[3s]** Quick scan for obvious error (signs, units, completion)
2. **[10s]** Test answer choice (C) in original problem
3. **[10s]** If (C) fails, identify trap pattern from choices
4. **[7s]** Apply most likely fix and select

If <30s remaining:

1. **[5s]** Eliminate trap answer(s) you found

2. **[5s]** Check remaining choices for reasonableness
3. **[5s]** Select most "competition-like" answer
4. **[5s]** Mark for review if time permits later

Common Quick Fixes:

- **Sign trap**: Multiply by -1
- **Incomplete trap**: Complete the final step
- **Unit trap**: Convert to requested units
- **Square/root trap**: Apply missing operation
- **Counting trap**: Check complement or add/subtract 1

4.7 Trap-to-Technique Quick Reference

Common Trap	**Related Technique**	**Quick Fix**
Non-monic polynomial in Vieta's	Vieta's (T16)	Divide by leading coefficient first
Point position in PoP	Power of Point (T23)	Check if inside/outside circle
Negative terms in AM-GM	AM-GM (T7)	Verify all terms ≥ 0
Wrong constant in SFFT	SFFT (T5)	$(x+a)(y+b) = ab + c$
Linearity for products	Linearity of Exp. (T11)	Only for sums, not products
Equality not achievable	Cauchy-Schwarz (T22)	Check proportionality feasible
Poor coordinate choice	Coordinate Conv. (T2)	Use symmetry point as origin
Angle bisector ratio	Angle Bisector (T10)	$\frac{BD}{DC} = \frac{AB}{AC}$ not $\frac{AB}{BC}$

4.8 Quick Reference: Top 20 Competition Traps

One-Page Trap Summary for Exam Day

Print this page for quick review before competitions!

Trap Name	Severity	Frequency	Quick Check
Early-Band Traps (Problems 1–10)			
Obvious Answer	H	25%	Did you finish the calculation?
Forgot to Finish	H	30%	Is this what's asked for?
Sign Error	M	20%	Check negative signs
Off-by-One	M	35%	Inclusive vs exclusive?
Wrong Operation	M	20%	Add vs multiply?
Units/Conversions	H	25%	Match units throughout
Successive Percent	M	15%	Multiply factors, don't add
Average Speed	M	20%	Use harmonic mean
Mid-Band Traps (Problems 11–20)			
Partial Solution	H	30%	Solved for x, asked for f(x)?
Special Case	M	25%	Test n=1, n=2
Overcounting	M	30%	Check for duplicates
Extraneous Solutions	H	25%	Verify in original
Absolute Value Cases	M	20%	Both positive and negative
Perm vs Combination	H	35%	Does order matter?
Scaling Confusion	M	25%	$L \to k$, $A \to k^2$, $V \to k^3$
Advanced Traps (Problems 15–20)			
Complex Conjugate	M	15%	Roots come in pairs
Trig Quadrant	H	25%	ASTC for signs
Functional Injectivity	M	20%	Verify 1-1 and onto
AM-GM Sign	M	20%	All terms ≥ 0?
Cauchy-Schwarz Equal	M	15%	Proportionality feasible?

Universal Quick Checks (10 seconds max):

- ✓ Re-read what's being asked
- ✓ Check answer magnitude reasonable
- ✓ Verify constraints satisfied
- ✓ Test in original if time permits
- ✓ Units match throughout
- ✓ Domain restrictions enforced
- ✓ Sign consistency maintained
- ✓ No division by zero
- ✓ Probabilities in [0,1]
- ✓ Counting results are integers

Emergency Trap Indicators in Answer Choices:

- Both x and -x present → sign trap likely
- Both x and x^2 present → incomplete calculation trap
- Answers differ by 1 → off-by-one trap
- Nice numbers in hard problem → trap answer included
- Your answer not listed → you fell for a trap

4.9 Section Summary: Trap Pattern Mastery

Key Takeaways for Trap Avoidance

Statistical Impact of Trap Awareness:

- Students aware of common traps improve accuracy by 15–20% on problems 1–10
- Trap recovery strategies can salvage 5–10 additional points per competition
- Time saved by recognizing traps early: 30–60 seconds per affected problem

Integration with Competition Strategy:

1. **Pre-Competition**: Review Quick Reference sheet (5 minutes)
2. **During Classification (0–15s)**: Scan answer choices for trap indicators
3. **During Execution (15–35s)**: Apply domain-specific trap awareness
4. **During Verification (50–65s)**: Use Enhanced Trap Detection Protocol
5. **If Stuck (35–50s)**: Check if you've fallen into a trap using Recovery Strategies

Most Important Traps by Problem Band:

- **Problems 1–5**: Units/Conversions, Obvious Answer, Forgot to Finish
- **Problems 6–10**: Off-by-One, Sign Error, Average Speed
- **Problems 11–15**: Partial Solution, Perm vs Combination, Extraneous Solutions
- **Problems 16–20**: Complex Conjugate, Trig Quadrant, Functional Injectivity

Remember: Trap awareness is not paranoia—it's strategic preparation. Most traps follow predictable patterns, and recognizing them gives you a significant competitive advantage.

5 Common Mistakes by Problem Type

Relationship to Trap Patterns

While Section 4 focused on **trap patterns** (deliberate answer choices designed to catch errors), this section addresses **common mistakes** (errors in technique or understanding). The distinction:

- **Trap Patterns**: External—placed by problem writers to catch predictable errors
- **Common Mistakes**: Internal—arise from misconceptions or careless execution
- **Overlap**: Many traps exploit common mistakes, so addressing both is essential

This section provides domain-specific mistake prevention strategies to complement the trap avoidance techniques.

How to Use This Section

During Practice:

- Review mistakes in your weak domains before problem sets
- Use the difficulty indicators to focus on your problem range
- Practice with the "Most Costly Mistakes" list for maximum improvement

During Competition:

- **Classification Phase (0–15s)**: Recall domain-specific pitfalls
- **Execution Phase (15–35s)**: Avoid computational mistakes
- **Verification Phase (50–65s)**: Use checklist to catch errors
- **Emergency Phase (65–120s)**: Check for most costly mistakes

Difficulty Indicators:

- 🟢 Early-band (Problems 1–10): Basic mistakes
- 🟡 Mid-band (Problems 11–15): Intermediate mistakes
- 🔴 Late-band (Problems 16–20): Advanced mistakes

The 10 Most Costly Mistakes (Highest Point Loss)

1. **Extraneous Solutions**: Not checking solutions in original equation (🟡🔴)
2. **Domain Restrictions**: Ignoring log/sqrt/denominator constraints (🟢🟡)
3. **Off-by-One Errors**: Fence-post, indexing, circular arrangements (🟢🟡)
4. **Permutation vs Combination**: Using wrong counting method (🟢)
5. **Scale Factor Confusion**: Linear vs area vs volume scaling (🟢🟡)
6. **Inequality Direction**: Not flipping when multiplying by negative (🟢)
7. **AM-GM Without Checking**: Applying to negative/mixed terms (🟡🔴)
8. **Modular Division**: Dividing without checking gcd (🟡🔴)
9. **Distribution Errors**: $(a+b)^2 \neq a^2 + b^2$, missing middle term (🟢)
10. **Not Answering Asked Question**: Solving for intermediate value (🟢🟡)

Statistics: Avoiding these 10 mistakes typically improves scores by 15–25% (3–5 problems).

5.1 Algebraic Problems - Common Mistakes

Algebra Pitfalls to Avoid

1. **Distributing Incorrectly** 🟢
   - **Mistake**: $(a+b)^2 = a^2 + b^2$ (forgetting the $2ab$ term)
   - **Correct**: $(a+b)^2 = a^2 + 2ab + b^2$
   - **Example**: $(3+4)^2 = 49 \neq 9 + 16 = 25$ (missing $2 \cdot 3 \cdot 4 = 24$)
   - **Prevention**: Always expand systematically or use FOIL; verify with small numbers
2. **Canceling Terms Incorrectly** 🟢

- **Mistake**: $\frac{x+3}{x+5} = \frac{3}{5}$ (canceling the x's)
- **Correct**: Cannot cancel terms, only factors
- **Prevention**: Factor completely before canceling

3. **Square Root Errors** ●●
   - **Mistake**: $\sqrt{x^2} = x$ (forgetting absolute value)
   - **Correct**: $\sqrt{x^2} = |x|$
   - **Prevention**: Consider sign of x when simplifying
4. **Logarithm Property Misuse** ●●
   - **Mistake**: $\log(a+b) = \log a + \log b$
   - **Correct**: $\log(ab) = \log a + \log b$
   - **Prevention**: Memorize correct properties; no property for log of sum
5. **Exponent Rule Confusion** ●
   - **Mistake**: $(x^a)^b = x^{a+b}$
   - **Correct**: $(x^a)^b = x^{ab}$
   - **Prevention**: Remember: powers multiply when raising a power to a power
6. **Inequality Direction Flip** ●●
   - **Mistake**: Multiplying/dividing an inequality by a negative without flipping the sign
   - **Correct**: Reverse the inequality direction when multiplying/dividing by a negative
   - **Prevention**: Note the sign of the multiplier/divisor before the operation
7. **Extraneous Solutions and Domain Restrictions** ●●
   - **Mistake**: Accepting all algebraic solutions after squaring or cross-multiplying; ignoring domain limits (logs/roots, zero denominators)
   - **Correct**: Check each candidate in the original equation; enforce domain: arguments of log positive, even roots of nonnegative, denominators nonzero
   - **Prevention**: Write domain first; always verify solutions in the original statement
8. **SFFT Constant Error** ●
   - **Mistake**: Wrong constant when completing rectangle in $xy + ax + by = c$
   - **Correct**: $(x+a)(y+b) = ab + c$ (not $ab - c$)
   - **Prevention**: Verify by expanding; see Technique 5 in Conversion guide
9. **Substitution Domain Loss** ●
   - **Mistake**: Not checking domain after back-substitution
   - **Correct**: If $t = x + \frac{1}{x}$, then $|t| \geq 2$ for real $x \neq 0$
   - **Prevention**: Track domain restrictions throughout; see Technique 1
10. **Complex Number Errors** ●●
    - **Mistake**: $|a+bi| = a+b$ or forgetting $i^2 = -1$ in expansions
    - **Correct**: $|a+bi| = \sqrt{a^2+b^2}$; always simplify i^2 to -1
    - **Example**: $|3+4i| = 5$ (not 7); $(1+i)^2 = 1 + 2i - 1 = 2i$ (not $1 + 2i + 1$)
    - **Prevention**: Use modulus formula; expand carefully with $i^2 = -1$
11. **Function Composition Order** ●
    - **Mistake**: $f(g(x)) = g(f(x))$ assumption; wrong evaluation order
    - **Correct**: $(f \circ g)(x) = f(g(x))$ means apply g first, then f
    - **Example**: If $f(x) = x^2$, $g(x) = x + 1$: $f(g(2)) = f(3) = 9$ but $g(f(2)) = g(4) = 5$
    - **Prevention**: Work inside-out; label intermediate steps

5.2 Geometric Problems - Common Mistakes

Geometry Pitfalls to Avoid

1. **Assuming Without Justification** ●●
   - **Mistake**: Assuming a triangle is right-angled when not stated
   - **Prevention**: Only use given information; check for special markers
2. **Confusing Radius and Diameter** ●

- **Mistake**: Using diameter in place of radius in formulas
- **Correct**: Area $= \pi r^2$, not πd^2
- **Prevention**: Always identify r vs d before calculating

3. **Incorrect Similar Triangle Ratios** ●●
   - **Mistake**: Setting up wrong correspondence between sides
   - **Prevention**: Label corresponding vertices clearly; match angles first
4. **3D Visualization Errors** ●●
   - **Mistake**: Confusing face diagonal with space diagonal
   - **Prevention**: Draw clear diagrams; use coordinate geometry when complex
5. **Arc Length vs Sector Area** ●●
   - **Mistake**: Using $r\theta$ for area or $\frac{1}{2}r^2\theta$ for arc length
   - **Correct**: Arc length $= r\theta$ (with θ in radians), Sector area $= \frac{1}{2}r^2\theta$
   - **Prevention**: Remember arc is 1D (linear), sector is 2D (area)
6. **Pythagorean Misuse** ●
   - **Mistake**: Applying $a^2 + b^2 = c^2$ in non-right triangles
   - **Correct**: Use Law of Cosines when the angle is not $90°$
   - **Prevention**: Confirm a right angle is given or implied before using Pythagorean Theorem
7. **Triangle Inequality Oversight** ●
   - **Mistake**: Assuming side lengths can form any triangle
   - **Correct**: Each side $<$ sum of other two; each side $>$ difference of other two
   - **Prevention**: Quick check before computing area/angles
8. **Scale Factor vs Area/Volume** ●●
   - **Mistake**: Scaling areas/volumes linearly with side scale factor k
   - **Correct**: Area scales by k^2; volume scales by k^3
   - **Prevention**: Identify whether a length, area, or volume is asked
9. **Angle Bisector Ratio Error** ●
   - **Mistake**: Using $\frac{BD}{DC} = \frac{AB}{BC}$ instead of $\frac{AB}{AC}$
   - **Correct**: Angle bisector theorem: $\frac{BD}{DC} = \frac{AB}{AC}$
   - **Prevention**: Remember: ratio involves the two sides forming the angle
   - **Reference**: See Technique 10 (Angle Bisector Theorem)
10. **Poor Coordinate Origin Choice** ●●
    - **Mistake**: Arbitrary origin making calculations complex
    - **Correct**: Place origin at symmetry point, centroid, or key vertex
    - **Prevention**: Exploit symmetry; see Technique 2 (Coordinate Conversion)
11. **Trigonometric Identity Errors** ●●
    - **Mistake**: $\sin(2x) = 2\sin(x)$ or $\cos(A + B) = \cos A + \cos B$
    - **Correct**: $\sin(2x) = 2\sin(x)\cos(x)$; $\cos(A + B) = \cos A \cos B - \sin A \sin B$
    - **Example**: $\sin(60°) = \frac{\sqrt{3}}{2} \neq 2\sin(30°) = 2 \cdot \frac{1}{2} = 1$
    - **Prevention**: Memorize key identities; verify with special angles
12. **Vector Calculation Mistakes** ●●
    - **Mistake**: $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$ (triangle inequality confusion)
    - **Correct**: $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ with equality only when parallel
    - **Example**: $(3, 0) + (0, 4)$ has magnitude 5, not 7
    - **Prevention**: Use component addition; apply Pythagorean theorem for magnitude

5.3 Counting/Probability Problems - Common Mistakes

Counting Pitfalls to Avoid

1. **Order vs No Order Confusion** ●
   - **Mistake**: Using permutations when combinations needed (or vice versa)
   - **Key Question**: Does order matter?
   - **Prevention**: "Arrange" → permutation, "Choose/Select" → combination
2. **Overcounting in Complex Scenarios** ●●
   - **Mistake**: Counting the same arrangement multiple times
   - **Example**: Circular arrangements counted as $n!$ instead of $(n-1)!$
   - **Prevention**: Identify symmetries and identical cases
3. **Mishandling "At Least" Problems** ●●
   - **Mistake**: Trying to count directly instead of using complement
   - **Better**: P(at least one) = 1 - P(none)
   - **Prevention**: Always consider complementary counting for "at least"
4. **Independent vs Dependent Events** ●●
   - **Mistake**: Multiplying probabilities when events are dependent
   - **Prevention**: Check if first outcome affects second; adjust denominator
5. **Conditional Probability Errors** ●●
   - **Mistake**: $P(A|B) = P(B|A)$ assumption
   - **Correct**: $P(A|B) = \frac{P(A\cap B)}{P(B)}$
   - **Prevention**: Use definition carefully; draw Venn diagram
6. **Inclusion–Exclusion Omitted** ●
   - **Mistake**: Adding counts of overlapping sets without subtracting intersections
   - **Correct**: $|A \cup B| = |A| + |B| - |A \cap B|$; extend to three sets as needed
   - **Prevention**: Draw Venn diagrams for overlapping conditions
7. **Identical Objects or Letters** ●●
   - **Mistake**: Using $n!$ when objects have repeats (e.g., letters in BALLOON)
   - **Correct**: Use $n!/(a!\,b!\,\ldots)$ dividing by factorials of identical counts
   - **Example**: BALLOON has 7 letters: B(1), A(1), L(2), O(2), N(1) → $\frac{7!}{2!\cdot 2!} = 1260$
   - **Prevention**: Count multiplicities before choosing a formula
8. **Bayes' Theorem Misapplication** ●●
   - **Mistake**: Confusing $P(A|B)$ with $P(B|A)$; wrong direction of conditioning
   - **Correct**: $P(A|B) = \frac{P(B|A)\cdot P(A)}{P(B)}$ using total probability for denominator
   - **Example**: Disease testing: $P(\text{disease}|\text{positive}) \neq P(\text{positive}|\text{disease})$
   - **Prevention**: Draw tree diagram; identify what's given vs what's asked
9. **Expected Value Non-linearity** ●●
   - **Mistake**: $E[XY] = E[X] \cdot E[Y]$ for dependent variables
   - **Correct**: Only true when X and Y are independent
   - **Prevention**: Check independence before multiplying expectations
10. **Geometric Probability Errors** ●
    - **Mistake**: Using perimeter ratios instead of area ratios for 2D regions
    - **Correct**: Probability = (favorable area)/(total area) for uniform distribution
    - **Example**: Point in circle of radius 2 being within radius 1: $P = \frac{\pi\cdot 1^2}{\pi\cdot 2^2} = \frac{1}{4}$ (not $\frac{1}{2}$)
    - **Prevention**: Match dimensions: length for 1D, area for 2D, volume for 3D

5.4 Number Theory Problems - Common Mistakes

Number Theory Pitfalls to Avoid

1. **Modular Arithmetic Sign Errors** ●●
   - **Mistake**: $-1 \equiv 1 \pmod 2$ confusion
   - **Correct**: $-1 \equiv 1 \pmod 2$ is true (both odd)
   - **Prevention**: Add modulus to negative numbers to get positive residue
2. **Forgetting GCD Conditions** ●●
   - **Mistake**: Assuming $ax \equiv b \pmod m$ always has solution
   - **Correct**: Solution exists iff $\gcd(a, m)|b$
   - **Prevention**: Check GCD condition before solving
3. **Prime vs Composite Confusion** ●
   - **Mistake**: Treating 1 as prime or forgetting 2 is prime
   - **Remember**: 1 is neither prime nor composite; 2 is the only even prime
   - **Prevention**: Memorize small primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29...
4. **Floor/Ceiling Function Errors** ●●
   - **Mistake**: $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ (not always true)
   - **Correct**: $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$
   - **Example**: $x = 0.7, y = 0.7$: $\lfloor 0.7 \rfloor + \lfloor 0.7 \rfloor = 0 + 0 = 0$ but $\lfloor 1.4 \rfloor = 1$
   - **When Equal**: Left equality when $\{x\} + \{y\} < 1$ (fractional parts sum to less than 1)
   - **Prevention**: Always check fractional parts; test boundary cases
5. **Divisibility Rule Misapplication** ●
   - **Mistake**: "Divisible by 6" means divisible by 2 OR 3
   - **Correct**: Must be divisible by 2 AND 3
   - **Prevention**: For composite divisors, check all prime factors
6. **Illegal Division in Congruences** ●●
   - **Mistake**: Cancelling a factor in $ax \equiv b \pmod m$ when $\gcd(a, m) \neq 1$
   - **Correct**: Divide by $d = \gcd(a, m)$ only if $d \mid b$; then reduce modulus to m/d
   - **Prevention**: Check gcd before cancelling; use modular inverses only when $\gcd(a, m) = 1$
7. **Totient/Fermat Misuse** ●●
   - **Mistake**: Applying $a^{\varphi(m)} \equiv 1 \pmod m$ or $a^{p-1} \equiv 1 \pmod p$ when a is not coprime to the modulus
   - **Correct**: Require $\gcd(a, m) = 1$ (or $p \nmid a$ for prime p)
   - **Prevention**: Check coprimality before reducing exponents modulo $\varphi(m)$ or $p - 1$
8. **Wilson's Theorem Misapplication** ●
   - **Mistake**: Using $(p - 1)! \equiv -1 \pmod p$ for composite p
   - **Correct**: Wilson's theorem only holds when p is prime
   - **Example**: $(4 - 1)! = 6 \equiv 2 \pmod 4$ (not -1)
   - **Prevention**: Verify primality before applying Wilson's theorem
9. **Diophantine Solution Counting** ●●
   - **Mistake**: Forgetting to check for positive/integer constraints
   - **Correct**: After finding general solution, enforce all given constraints
   - **Example**: $x + y = 10$ has 11 non-negative integer solutions but only 9 positive ones
   - **Prevention**: List constraints explicitly; check boundary cases
10. **Quadratic Residue Errors** ●
    - **Mistake**: Assuming $x^2 \equiv a \pmod p$ always has solutions
    - **Correct**: Solutions exist iff a is a quadratic residue mod p
    - **Prevention**: Use Legendre symbol or check $a^{(p-1)/2} \equiv 1 \pmod p$

5.5 Sequence/Series Problems - Common Mistakes

Sequence Pitfalls to Avoid

1. **Off-by-One in Formulas** ●●
   - **Mistake**: Using $a_n = a_1 + nd$ instead of $a_n = a_1 + (n-1)d$
   - **Prevention**: Check with small values: a_1 should equal a_1, not $a_1 + d$
2. **Geometric Series Convergence** ●
   - **Mistake**: Applying $\frac{a}{1-r}$ when $|r| \geq 1$
   - **Correct**: Infinite geometric series converges only when $|r| < 1$
   - **Prevention**: Always check convergence condition first
3. **Telescoping Misidentification** ●●
   - **Mistake**: Assuming partial fractions always telescope
   - **Prevention**: Write out first few terms to verify cancellation pattern
4. **Sum Formula Bounds** ●
   - **Mistake**: $\sum_{k=1}^{n} k = \frac{n(n-1)}{2}$ (wrong formula)
   - **Correct**: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$
   - **Prevention**: Memorize key formulas; verify with $n = 1, 2, 3$
5. **Recursive Definition Errors** ●
   - **Mistake**: Starting recursion at wrong index
   - **Prevention**: Clearly identify initial conditions and first recursive step
6. **Geometric Term Indexing** ●●
   - **Mistake**: $a_n = a_1 r^n$ instead of $a_n = a_1 r^{n-1}$
   - **Prevention**: Verify with $n = 1$ gives a_1
7. **Summation Index Shifts** ●●
   - **Mistake**: Changing indices without adjusting bounds/terms correctly
   - **Prevention**: Write out first few terms before and after the shift to match terms

5.6 Inequality and Optimization Problems - Common Mistakes

Inequality and Optimization Pitfalls to Avoid

1. **AM-GM Without Checking Non-negativity** ●●
   - **Mistake**: Applying AM-GM to negative or mixed-sign terms
   - **Correct**: All terms must be non-negative for AM-GM
   - **Prevention**: Verify $x > 0$ before using $x + \frac{c}{x} \geq 2\sqrt{c}$
   - **Reference**: See Technique 7 (AM-GM Inequality)
2. **Cauchy-Schwarz Direction Error** ●●
   - **Mistake**: Using CS to minimize instead of maximize (or vice versa)
   - **Correct**: CS gives upper bound: $(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$
   - **Prevention**: Remember CS provides upper bounds for dot products
   - **Reference**: See Technique 22 (Cauchy-Schwarz)
3. **Rearrangement Inequality Confusion** ●
   - **Mistake**: Same order for minimum, reverse for maximum
   - **Correct**: Same order maximizes, reverse order minimizes
   - **Prevention**: Think "like with like" for maximum value
   - **Reference**: See Technique 15 (Rearrangement Inequality)
4. **Ignoring Equality Conditions** ●●
   - **Mistake**: Assuming equality is achievable without checking
   - **Correct**: Verify equality conditions satisfy all constraints
   - **Prevention**: For AM-GM: all equal; for CS: proportional; check feasibility
5. **Weighted AM-GM Errors** ●
   - **Mistake**: Incorrect weights or forgetting to normalize

- **Correct**: Weights must sum to 1 or use integer repetitions
- **Prevention**: Double-check weight calculations and applications

5.7 Mixed Domain Problems - Common Mistakes

Mixed Domain Pitfalls to Avoid

1. **Algebra-Geometry Interface** ●●
   - **Mistake**: Setting up coordinates without considering constraints
   - **Correct**: Ensure algebraic equations reflect all geometric conditions
   - **Example**: Circle tangent to line requires distance = radius, not just intersection
   - **Prevention**: List all constraints before algebraicizing
2. **Counting with Geometric Constraints** ●
   - **Mistake**: Double-counting symmetric configurations
   - **Correct**: Identify symmetry group; divide by symmetry factor
   - **Example**: Coloring vertices of square: divide by 8 for rotations and reflections
   - **Prevention**: Use Burnside's lemma or count fundamental regions
3. **Number Theory in Counting** ●●
   - **Mistake**: Ignoring divisibility constraints in enumeration
   - **Correct**: Use modular arithmetic to restrict valid cases
   - **Example**: Count multiples of 3 in range: use $\lfloor b/3 \rfloor - \lfloor (a-1)/3 \rfloor$
   - **Prevention**: Translate constraints to congruences first
4. **Probability with Number Theory** ●
   - **Mistake**: Assuming uniform distribution for number-theoretic properties
   - **Correct**: Consider density of primes, coprime pairs, etc.
   - **Example**: Probability two random integers are coprime $= \frac{6}{\pi^2}$ (not $\frac{1}{2}$)
   - **Prevention**: Use known densities or compute explicitly for small ranges
5. **Optimization with Discrete Constraints** ●
   - **Mistake**: Using calculus on integer-only variables
   - **Correct**: Check integer points near continuous optimum
   - **Example**: Minimize $x^2 + (10/x)^2$ for integer x: check $x = 2, 3$ near $\sqrt[4]{10}$
   - **Prevention**: Find continuous solution, then test nearby integers

5.8 Quick Mistake-Prevention Checklist

Universal Mistake Prevention - Timeline Integrated

Classification Phase [0–15s]: Jump to Classification
- ☐ Read problem completely - what's actually being asked?
- ☐ Note special conditions (positive, integer, distinct, etc.)
- ☐ Identify problem type to anticipate domain-specific errors
- ☐ Check answer choices for trap indicators (see Section 4)

Execution Phase [15–35s]: Jump to Domain Lanes
- ☐ Write neatly - messy work leads to errors
- ☐ Check algebra at each step - don't compound errors
- ☐ Question assumptions - are you adding unstated conditions?
- ☐ Apply domain-specific prevention strategies from this section

Pivot Phase [35–50s]: Jump to Pivot Strategy
- ☐ If stuck, check for common mistakes in your approach
- ☐ Consider mixed-domain errors if problem combines areas
- ☐ Try simpler cases to verify method correctness

Verification Phase [50–65s]: Jump to Answer Testing

- ☐ Verify answer satisfies all original conditions
- ☐ Check if answer is reasonable (magnitude, sign, type)
- ☐ Ensure you've answered what was asked (not intermediate step)
- ☐ Test in original equation if time permits

Common Red Flags by Problem Band:

- • **Early (1–10)**: Too complex → you're overthinking
- • **Mid (11–15)**: Too clean → check for traps
- • **Late (16–20)**: Uses subset of given info → incomplete

Quick Reference Table: Problem Number to Common Mistakes

Problem Range	Most Common Mistakes	Prevention Priority
1–5	Distribution errors, Units, Sign errors	Basic algebra checks
6–10	Off-by-one, Perm vs Comb, Domain loss	Careful counting
11–15	Extraneous solutions, Scale factors, Floor/ceiling	Verify all solutions
16–20	AM-GM conditions, Modular division, Complex numbers	Check applicability

6 Mental Math Techniques for Speed

Integration with Rapid Classification Timeline

This section is fully aligned with the Rapid Problem Classification System (Section 1)
Mental Math Timeline for Every Problem:

- **0–15s**: Pattern recognition mental math (15-Second Classification)
- **15–35s**: Domain-specific mental calculations (Domain Lanes)
- **35–50s**: Quick estimation for pivots (35-Second Pivot)
- **50–65s**: Answer verification calculations (Answer Testing)
- **65–120s**: Emergency estimation (Red Flags)

Key Principle: Mental math is not just calculation—it's pattern recognition, estimation, and verification integrated into your classification timeline.

6.1 15-Second Mental Math for Classification [0–15s]

0–10 s **WHO/WHEN Framework for Mental Math**

Classification Phase Mental Math [0–15s]

0–15 s **Instant Recognition Calculations**
WHO: Problems requiring immediate numerical insight during classification **WHEN**: Apply within first 4–10 seconds to identify technique (includes key pattern signals formerly labeled WHERE)
Priority Mental Checks [0–4s]:

- **Magnitude estimation**: Is answer in hundreds? thousands? Compare to choices
- **Divisibility quick test**: Check if numbers suggest modular arithmetic
- **Perfect square/cube recognition**: $169=13^2$, $225=15^2$, $343=7^3$
- **Fraction to decimal**: $\frac{1}{3} \approx 0.333$, $\frac{1}{6} \approx 0.167$, $\frac{1}{7} \approx 0.143$, $\frac{1}{8} = 0.125$, $\frac{1}{9} \approx 0.111$

Pattern Triggers [4–10s]:

- See 1001? → Think $7\times11\times13$ (useful for divisibility)
- See differences of squares? → Factor instantly: 144-25 = (12+5)(12-5) = 17×7
- Quadratic coefficients? → Mental discriminant: b^2-4ac for real solutions
- Geometric sequence? → Ratio check: 3,12,48 → r=4

Answer Choice Intelligence [10–15s]:

- Evenly spaced → Middle value often correct (test C first)
- Powers of 2 → Check if you completed all doublings
- Mix of fractions/decimals → Convert mentally to compare
- One outlier → Either trap or correct answer

6.2 20-Second Domain-Specific Mental Math [15–35s]

15–35 s **Execution Phase Calculations**

Algebra Lane Mental Math [15–35s]

15–35 s Tag: ALG
WHO: Algebraic manipulations requiring speed **WHERE**: Equations, factoring, function evaluation
WHEN: During domain execution after classification
[10–15s] Immediate Mental Actions:

- **Vieta's mental**: For x^2+bx+c, sum=-b, product=c (adjust for non-monic)
- **SFFT mental**: xy+3x+5y=7 → (x+5)(y+3)=22 → factor 22: 1×22, 2×11
- **Completing square**: $x^2+6x \to (x+3)^2-9$ (half coefficient, square it)

[15–20s] Pattern Matching:

- **Difference of squares**: a^2-b^2 = (a+b)(a-b) instant factoring
- **Sum/difference of cubes**: 27+8 = 3^3+2^3 = (3+2)(9-6+4) = 5×7
- **Hidden quadratic**: If x+1/x=3, then x^2+1/x^2=9-2=7

[20–25s] Quick Substitution Check:

- Test x=0,1,-1,2 mentally before complex solving
- For inequalities: check boundary values
- Domain restrictions: denominators≠ 0, logs> 0

Geometry Lane Mental Math [15–35s]

15–35 s Tag: GEO

WHO: Geometry problems needing quick calculations **WHERE**: Triangle ratios, circle calculations, coordinate geometry **WHEN**: During synthetic or coordinate approach

[10–15s] Pythagorean Triple Recognition:

- Common: (3,4,5), (5,12,13), (8,15,17), (7,24,25)
- Scaled: 6,8,10 = 2×(3,4,5); 10,24,26 = 2×(5,12,13)
- Check: Does a^2+b^2=c^2? 144+25=169? Yes → 12^2+5^2=13^2

[15–20s] Special Angle Values:

- $30° : sin = \frac{1}{2}, cos = \frac{\sqrt{3}}{2}, tan = \frac{1}{\sqrt{3}}$
- $45° : sin = cos = \frac{1}{\sqrt{2}}, tan = 1$
- $60° : sin = \frac{\sqrt{3}}{2}, cos = \frac{1}{2}, tan = \sqrt{3}$

[20–25s] Area/Volume Scaling:

- Linear scale k → Area scales k^2 → Volume scales k^3
- Circle: $A = \pi r^2$, double radius → 4× area
- Similar triangles: ratio of areas = (ratio of sides)^2

Counting/NT/Probability Mental Math [15–35s]

15–35 s Tags: CNT/NT/PROB

Counting Quick Calculations:

- **Small factorials**: $5! = 120, 6! = 720, 7! = 5040$
- **Choose formulas**: C(n,2)=n(n-1)/2, C(n,3)=n(n-1)(n-2)/6
- **Stars and bars**: n identical into k boxes = C(n+k-1,k-1)

Number Theory Modular Arithmetic:

- **Quick mods**: $1000 \equiv -1(mod 7), 100 \equiv 2(mod 7), 10 \equiv 3(mod 7)$
- **Fermat's**: $2^6 \equiv 1(mod 7)$, so $2^{100} = 2^{96} \cdot 2^4 \equiv 16 \equiv 2(mod 7)$
- **Digital root**: Sum digits repeatedly → result mod 9

Probability Fractions:

- Always reduce: $\frac{12}{30} = \frac{2}{5}$ immediately
- Check $0 \leq P \leq 1$ (common trap)
- Complement: P(at least 1) = 1 - P(none)

6.3 15-Second Pivot Strategy Mental Math [35–50s]

35–50 s **Quick Calculations for Strategy Changes**

35-Second Pivot Mental Math

35–50 s **Emergency Estimation**

WHO: Problems where primary technique isn't working **WHERE**: Stuck calculations, complex

expressions, time pressure **WHEN**: At 35s checkpoint when no progress visible

Universal Pivot Calculations:

- **Constraint elimination first**: Apply 1-2 bounds before choice testing
- **Magnitude bounds**: Eliminate answers differing by 10× or more
- **Parity check**: Even/odd elimination
- **Mod check**: Quick mod 2,3,5 to eliminate choices

Domain-Specific Pivots:

- **Algebra stuck**: Try x=1,2,3 or test answer C
- **Geometry stuck**: Estimate using $\pi \approx 3, \sqrt{2} \approx 1.4, \sqrt{3} \approx 1.7$
- **Counting stuck**: Calculate n=1,2,3 cases, find pattern
- **NT stuck**: Check small moduli (2,3,5,7)

Quick Approximations:

- $\pi \approx 3.14159$ (Note: $\frac{22}{7} \approx 3.14286$ is slightly larger)
- $e \approx 2.71828$
- $\sqrt{2} \approx 1.41421$
- $\sqrt{3} \approx 1.73205$
- $\sqrt{5} \approx 2.23607$
- Golden ratio $\phi = \frac{1+\sqrt{5}}{2} \approx 1.61803$

6.4 15-Second Answer Testing Mental Math [50–65s]

50–65 s **Verification Phase Calculations**

50-Second Answer Verification

50–65 s **Quick Testing Protocol**

WHO: All problems reaching 45s mark **WHERE**: Multiple choice verification, elimination **WHEN**: Time pressure or uncertain about solution

Testing Priority [45–50s]:

1. Apply constraint elimination first - test promising candidate after narrowing
2. If C too large → test B; if C too small → test D
3. For optimization: test extremes (A) and (E)
4. Test "nice" numbers: 0, 1, -1, perfect squares

Rapid Elimination [50–55s]:

- **Units digit**: Multiply units only for quick check
- **Magnitude**: One decimal place accuracy often enough
- **Sign**: Positive/negative elimination
- **Domain**: Length$> 0, probability \in [0, 1], count \geq 0$

Pattern Recognition [60–65s]:

- Arithmetic progression → middle term likely
- Geometric progression → check ratio consistency
- One outlier → trap or correct answer
- Similar pairs → one is trap, one correct

6.5 55-Second Emergency Mental Math [65–120s]

65–120 s **Bailout Calculations**

Red Flag Mental Math

60–120 s **Last Resort Estimation**

WHO: Problems in danger zone **WHERE**: No clear path after 60 seconds **WHEN**: Must make educated guess or skip

65-Second Decision Protocol:
- Eliminated 0 choices → Skip (blank = 1.5 pts)
- Eliminated 1 choice → Break-even (guess from 4)
- Eliminated 2+ choices → Always guess (EV > 2 pts)

Quick Elimination Tricks:
- **Dimensional analysis**: Area→squared, Volume→cubed
- **Extreme cases**: $n = 0, n = \infty, degeneratecases$
- **Symmetry**: Answer should respect problem symmetry
- **Conservation**: Total must equal sum of parts

Strategic Guessing:
- Problems 1-5: Trust simple answers
- Problems 6-10: Middle complexity answers
- Problems 11-15: Avoid too simple
- Problems 16-20: Complex answers more likely

6.6 Core Calculation Techniques

Order of Magnitude and Estimation Techniques

Order of Magnitude Estimation:
- Round to nearest power of 10 for quick bounds
- Example: $387 \times 52 \approx 400 \times 50 = 20,000$
- For division: $\frac{8947}{293} \approx \frac{9000}{300} = 30$
- Scientific notation: $3.7 \times 10^4 \times 2.1 \times 10^{-2} \approx 4 \times 2 \times 10^2 = 800$

Significant Figures Rounding:
- Keep 2-3 significant figures for mental math
- $\sqrt{87} \approx \sqrt{90} \approx 9.5$ (since $9^2 = 81, 10^2 = 100$)
- Linear interpolation: $\sqrt{87} \approx 9 + \frac{87-81}{100-81} \approx 9.3$

Fermi Estimation Principles:
- Break complex problems into simpler estimates
- Use known benchmarks (population of US $\approx 3 \times 10^8$)
- Combine upper and lower bounds for range
- Geometric mean of bounds often good estimate: $\sqrt{lower \times upper}$

Percentage Error Quick Check:
- If estimate is within 10%, usually sufficient for elimination
- Error $= \frac{|actual - estimate|}{actual} \times 100\%$
- For small changes: $(1 + x)^n \approx 1 + nx$ when x is small

Multiplication Shortcuts (High-Probability Calculations)

- **×5**: Divide by 2, then ×10 (e.g., $84 \times 5 = 42 \times 10 = 420$)
- **×9**: Multiply by 10, subtract original (e.g., $73 \times 9 = 730 - 73 = 657$)
- **×11 (2-digit)**: Add digits and insert; if sum ≥10, carry the 1 (e.g., $53 \times 11 = 583$, $87 \times 11 = 957$)
- **×11 (3+ digits)**: From right to left: write last digit; for interior digits write neighbor sums with carries; finish with first digit
- **×15**: Multiply by 10, add half of that (e.g., $24 \times 15 = 240 + 120 = 360$)
- **×25**: Divide by 4, then ×100 (e.g., $88 \times 25 = 22 \times 100 = 2200$)
- **×125**: Divide by 8, then ×1000 (e.g., $64 \times 125 = 8 \times 1000 = 8000$)

- **Near-square**: $(n \pm d)^2 = n^2 \pm 2nd + d^2$ (use a nearby square)
- **Near-hundred products**: $(100 + a)(100 + b) = 10000 + 100(a + b) + ab$ (similarly for 1000)
- **Difference of squares**: $a^2 - b^2 = (a + b)(a - b)$ (instant factoring)

Division Tricks

- **÷5**: Multiply by 2, then ÷10 (e.g., $73 \div 5 = 146 \div 10 = 14.6$)
- **÷25**: Multiply by 4, then ÷100
- **Divisible by 6**: Divisible by both 2 and 3
- **Divisibility by 2**: Last digit even
- **Divisibility by 3**: Sum of digits divisible by 3
- **Divisibility by 4**: Last two digits divisible by 4
- **Divisibility by 5**: Last digit 0 or 5
- **Divisibility by 8**: Last three digits divisible by 8
- **Divisibility by 9**: Sum of digits divisible by 9
- **Divisibility by 11**: Alternating sum of digits = 0 or multiple of 11
- **Tip for 7 (quick mod)**: Reduce using $10 \equiv 3 \pmod 7$ or $100 \equiv 2 \pmod 7$ for mental checks

Percentage Calculations

- **10%**: Move decimal left once
- **5%**: Half of 10%
- **15%**: 10% + 5%
- **20%**: Divide by 5
- **25%**: Divide by 4
- **1%**: Divide by 100 (move decimal left twice)
- **2.5%**: Divide by 40
- **12.5%**: Divide by 8
- **62.5%**: Divide by 1.6 (or $\times \frac{5}{8}$)
- **$33\frac{1}{3}$%**: Divide by 3
- **50%**: Divide by 2

Common Squares and Cubes

- Memorize: 1^2–25^2, especially $11^2 = 121$, $12^2 = 144$, $13^2 = 169$, $15^2 = 225$
- Memorize: $2^3 = 8$, $3^3 = 27$, $4^3 = 64$, $5^3 = 125$, $6^3 = 216$
- Pattern: $(10a + 5)^2 = 100a(a + 1) + 25$ (e.g., $35^2 = 1225$)

Powers and Logarithms Quick Rules

- **Memorize powers (high yield)**: 2^1–2^{10}: $2, 4, 8, 16, 32, 64, 128, 256, 512, 1024$
- Memorize: 3^1–3^6: $3, 9, 27, 81, 243, 729$; 5^1–5^6: $5, 25, 125, 625, 3125, 15625$
- **Log rules**: $\log(ab) = \log a + \log b$, $\log\left(\frac{a}{b}\right) = \log a - \log b$, $\log(a^k) = k \log a$
- **Change of base**: $\log_b a = \frac{\log_c a}{\log_c b}$ (use base 10 or 2 for speed)
- **Common log values**: $\log_{10} 2 \approx 0.301$, $\log_{10} 3 \approx 0.477$, $\log_{10} 5 = 0.699$, $\log_{10} 7 \approx 0.845$
- **Natural log values**: $\ln 2 \approx 0.693$, $\ln 3 \approx 1.099$, $\ln 10 \approx 2.303$

6.7 High-Probability Technique Mental Math [Problems 11–20]

High-Priority Mental Calculations (Aligned with Companion Document)

Reference: These align with High-Probability Techniques in AMC12_COMC_Conversion_Techniques.tex

1. **Algebraic Substitution (T1):**
   - Mental substitution: $t = x + \frac{1}{x}$ implies $t^2 = x^2 + \frac{1}{x^2} + 2$
   - For $t = x + \frac{1}{x}$ with real $x \neq 0$: $|t| \geq 2$
   - Quick check: If $t = 3$, then $x^2 - 3x + 1 = 0$, so $x = \frac{3 \pm \sqrt{5}}{2}$
2. **Completing the Square (T8):**
   - Mental formula: $x^2 + bx = (x + \frac{b}{2})^2 - \frac{b^2}{4}$
   - Quick vertex: For $ax^2 + bx + c$, vertex at $x = -\frac{b}{2a}$
   - Circle form: $x^2 + y^2 + Dx + Ey + F = 0$ has center $(-\frac{D}{2}, -\frac{E}{2})$
3. **Vieta's Formulas (T16 - Now High-Priority):**
   - For $ax^2 + bx + c = 0$: sum $= -\frac{b}{a}$, product $= \frac{c}{a}$
   - Mental check: If sum = 5, product = 6, then roots are 2 and 3
   - Powers of roots: $r_1^2 + r_2^2 = (r_1 + r_2)^2 - 2r_1r_2$
4. **SFFT - Simon's Favorite Factoring Trick (T5):**
   - Mental conversion: $xy + ax + by = c$ becomes $(x + b)(y + a) = c + ab$
   - Example: $xy + 3x + 5y = 7 \to (x + 5)(y + 3) = 22$
   - Factor 22 mentally: $1 \times 22, 2 \times 11 \to$ solutions $(x, y) = (-4, 19), (-3, 8)$
5. **AM-GM Inequality (T7):**
   - For positive a, b: $\frac{a+b}{2} \geq \sqrt{ab}$ with equality when $a = b$
   - Mental application: $x + \frac{4}{x} \geq 2\sqrt{4} = 4$ for $x > 0$
   - Three variables: $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$
6. **Power of a Point (T23 - Now High-Priority):**
   - Internal point: $PA \cdot PB = PC \cdot PD$ for chords through P
   - External point: $PT^2 = PA \cdot PB$ for tangent PT and secant PAB
   - Mental check: If PT = 6 and PA = 4, then PB = 9
7. **Angle Bisector Theorem (T10):**
   - Ratio formula: $\frac{BD}{DC} = \frac{AB}{AC}$ (NOT $\frac{AB}{BC}$)
   - If AB = 6, AC = 8, BC = 10, then BD:DC = 3:4
   - Total BD + DC = BC to find exact lengths
8. **Inclusion-Exclusion (T21 - Now High-Priority):**
   - Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$
   - Three sets: Add singles, subtract pairs, add triple
   - Mental check: If $|A| = 30, |B| = 25, |A \cap B| = 10, then |A \cup B| = 45$

Additional High-Yield Mental Techniques

Complex Numbers:

- Powers of i: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$ (cycle repeats every 4)
- For i^n: find n mod 4 and use the pattern above
- $(a + bi)(a - bi) = a^2 + b^2$ (multiply by conjugate gives real number)
- Modulus: $|a + bi| = \sqrt{a^2 + b^2}$
- De Moivre's: $(r\text{cis}\theta)^n = r^n\text{cis}(n\theta)$ where $\text{cis}\,\theta = \cos\theta + i\sin\theta$
- Roots of unity: $\omega = e^{2\pi i/n}$ satisfies $\omega^n = 1$, and $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$
- Quick multiplication: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$

Function Composition:

- If $f(x) = ax + b$ with $a \neq 0$, then $f^{-1}(x) = \frac{x-b}{a}$
- $(f \circ f^{-1})(x) = x$ (always)

Sequence Shortcuts:

- Arithmetic: middle term = average of neighbors
- Geometric: middle term = geometric mean of neighbors
- Sum of first n integers: $\frac{n(n+1)}{2}$
- Sum of first n squares: $\frac{n(n+1)(2n+1)}{6}$
- Sum of first n cubes: $[\frac{n(n+1)}{2}]^2$

Series Sum Formulas:

- Arithmetic series: $S_n = \frac{n(a_1+a_n)}{2} = \frac{n[2a_1+(n-1)d]}{2}$
- Geometric series: $S_n = a\frac{1-r^n}{1-r}$ (for $r \neq 1$)
- Infinite geometric: $S = \frac{a}{1-r}$ (for $|r| < 1$)
- Telescoping series: Most terms cancel, leaving first and last

Binomial Expansion Patterns:

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$
- Pascal's triangle for coefficients: 1, 1-1, 1-2-1, 1-3-3-1, 1-4-6-4-1
- Binomial coefficient: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Linearity of Expectation (T11 - High-Priority):

- $E[X + Y] = E[X] + E[Y]$ always (no independence needed)
- For indicators: $E[\sum I_i] = \sum P(I_i = 1)$
- Example: Expected heads in 10 flips = $10 \times 0.5 = 5$

Cauchy-Schwarz (T22 - Now High-Priority):

- $(\sum a_ib_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$
- Equality when $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots$
- Mental: $(3 \times 4 + 5 \times 12)^2 \leq (9 + 25)(16 + 144) = 34 \times 160$

Number Theory Mental Math

Modular Arithmetic Patterns:

- Fermat's Little Theorem: $a^{p-1} \equiv 1 \pmod{p}$ for prime p, $\gcd(a, p) = 1$
- Euler's: $a^{\phi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$
- Quick ϕ: $\phi(p^k) = p^{k-1}(p - 1)$; $\phi(pq) = (p - 1)(q - 1)$ for distinct primes
- Chinese Remainder: Break into prime powers, solve separately, combine

Digital Root and Base Conversions (T9):

- Digital root: For $n > 0$, use $1 + ((n - 1) \bmod 9)$. Result is 1-9 (never 0)
- Special case: When $n \equiv 0 \pmod 9$, digital root = 9 (not 0)
- Base b: $N \equiv (\text{sum of digits}) \pmod{b - 1}$
- Last digit in base b: $N \bmod b$

- Divisibility by 11: alternating sum of digits

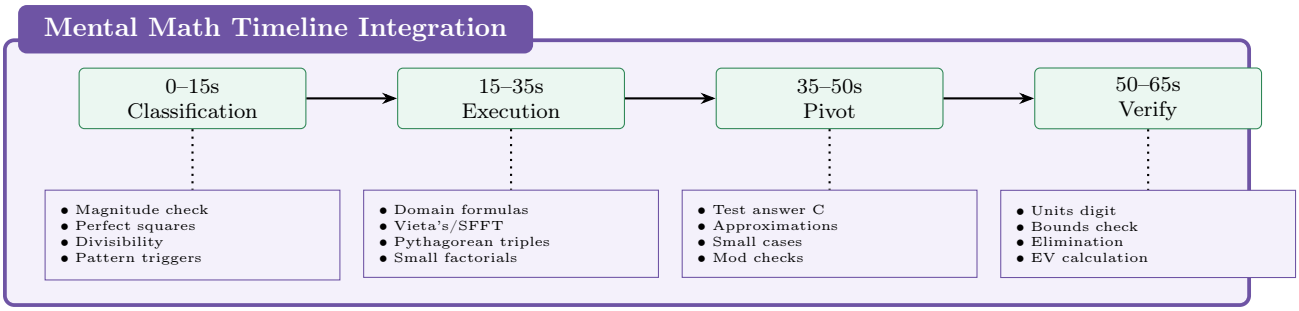
Trigonometric Values and Formulas

- **Special angles**: $30°, 45°, 60°$ values (memorized)
- **Complementary angles**: $\sin(\theta) = \cos(90° - \theta)$
- **Pythagorean triples**:
  - Common: (3,4,5), (5,12,13), (8,15,17), (7,24,25)
  - Additional: (9,40,41), (11,60,61), (13,84,85), (20,21,29)
  - Pattern: $(2n+1, 2n^2+2n, 2n^2+2n+1)$ for any n
  - Scaling: Any triple (a, b, c) gives (ka, kb, kc) for integer k
- **Sum-to-product formulas**:
  - $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$
  - $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$
- **Product-to-sum formulas**:
  - $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$
  - $\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$
- **Sum and difference identities**:
  - $\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$
  - $\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$
  - $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$
- **Double-angle**:
  - $\sin(2\theta) = 2\sin\theta\cos\theta$
  - $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$
- **Half-angle formulas**:
  - $\sin(\theta/2) = \pm\sqrt{\frac{1-\cos\theta}{2}}$
  - $\cos(\theta/2) = \pm\sqrt{\frac{1+\cos\theta}{2}}$
  - $\tan(\theta/2) = \frac{1-\cos\theta}{\sin\theta} = \frac{\sin\theta}{1+\cos\theta}$
  - Sign in half-angle formulas depends on the quadrant of $\theta/2$

Number Theory Shortcuts

- **Modular arithmetic**: Work mod small primes (2,3,5,7)
- **Fermat's Little**: If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$; in general $a^p \equiv a \pmod{p}$
- **Modular inverse**:
  - $a^{-1} \pmod{m}$ exists iff $\gcd(a, m) = 1$
  - For prime p: $a^{-1} \equiv a^{p-2} \pmod{p}$ (by Fermat's Little Theorem)
  - Extended Euclidean Algorithm finds $ax + my = 1$, then $x \equiv a^{-1} \pmod{m}$
  - Example: $3^{-1} \equiv 3^5 \equiv 243 \equiv 5 \pmod{7}$
- **Chinese Remainder Theorem**: Break into prime power components
- **Digital root**: For $n > 0$, use $1 + ((n-1) \bmod 9)$ which gives 1-9 (never 0). When $n \equiv 0 \pmod{9}$, result is 9. For $n = 0$, digital root $= 0$

6.8 Visual Workflow: Mental Math Decision Tree



6.9 Speed Calculation Drills

Timeline-Aligned Daily Practice

Classification Phase Drills [0–15s] - Do 3 daily:

1. Identify problem domain from keywords (3 seconds)
2. Match pattern to technique (WHO/WHEN) (3 seconds)
3. Analyze answer choices for clues (4 seconds)

Execution Phase Drills [15–35s] - Do 5 daily:

1. Apply Vieta's formulas mentally to find sum/product
2. Factor using SFFT: $xy + ax + by = c$
3. Complete the square for vertex form
4. Recognize Pythagorean triples
5. Calculate C(n,k) for small values

Pivot/Verification Drills [35–65s] - Do 3 daily:

1. Apply constraint elimination in 10 seconds
2. Eliminate 2-3 choices by magnitude
3. Check answer using mod arithmetic

Pattern Recognition Speed Drills:

1. Spot AM-GM opportunities (nonnegative with equality condition)
2. Identify Power of Point configurations
3. Recognize inclusion-exclusion setups
4. Find hidden substitutions ($t = x + \frac{1}{x}$)

6.10 Competition-Day Mental Prep

Pre-Competition Warmup (10 minutes)

- Review special values: $\pi \approx 3.14159$, $e \approx 2.71828$, $\sqrt{2} \approx 1.41421$, $\sqrt{3} \approx 1.73205$, $\phi \approx 1.61803$
- Practice 5 quick calculations without paper (factoring, Vieta's, modular arithmetic)
- Recall 3 favorite problem patterns (SFFT, completing square, inclusion-exclusion)
- Review trap answer indicators (outliers, too simple for problem number)

During Competition:

- **First 10 problems**: Trust mental math, minimize writing
- **Problems 11–20**: Mental for simple steps, write complex
- **Checking**: Mental estimation for magnitude verification
- **Time crunch**: Mental elimination of impossible answers (aim to kill 2–3 choices)

QUICK REFERENCE: Mental Math by Timeline Phase

Print This Mental Math Checklist

Classification [0–15s]:
- Magnitude estimation (hundreds? thousands?)
- Perfect square/cube recognition (169, 225, 343)
- Divisibility triggers ($1001 = 7\times11\times13$)
- Answer pattern analysis (test C first if evenly spaced)

Execution [15–35s]:
- **ALG**: Vieta's ($sum = -\frac{b}{a}, prod = \frac{c}{a}$), SFFT ($(x+b)(y+a) = c+ab$)
- **GEO**: Pythagorean triples, special angles ($30°, 45°, 60°$)
- **CNT**: Small factorials ($5! = 120, 6! = 720$), $C(n,2) = n(n-1)/2$
- **NT**: Fermat's ($a^{p-1} \equiv 1 mod p$), digital root
- **PROB**: Reduce fractions, check $0 \le P \le 1$

Pivot [35–50s]:
- Test answer C immediately
- Use approximations: $\pi \approx 3.14, \sqrt{2} \approx 1.41, e \approx 2.72$
- Calculate n=1,2,3 to find pattern
- Quick mod 2,3,5,7 checks

Verification [50–65s]:
- Units digit multiplication
- Magnitude bounds ($10\times$ elimination)
- Domain constraints (positive lengths, $P \in [0,1]$)
- Expected value for guessing

Emergency [65–120s]:
- Dimensional analysis (area→squared)
- Extreme cases ($n = 0, n = \infty$)
- Strategic guessing based on problem number
- Skip if eliminated < 2 choices

Key Techniques Cross-Reference: T1: Algebraic Substitution | T5: SFFT | T7: AM-GM | T8: Completing Square
T10: Angle Bisector | T11: Linearity of E | T16: Vieta's | T22: Cauchy-Schwarz
T23: Power of Point | T21: Inclusion-Exclusion | T9: Base/Digit Conversions

7 Post-Run Analysis for Improvement

7.1 Rapid Classification System Performance Analysis

Timeline Phase Performance Tracking

Phase-by-Phase Analysis (aligned with Section 1):

Phase	Target	Actual	Δ	Analysis
0–15s: Classification				
Domain Scan (0–4s)	4s			
Pattern Match (4–10s)	6s			
Answer Intelligence (10–15s)	5s			
15–35s: Execution				
Domain Lane Entry	5s			
Technique Application	15s			
35–50s: Pivot				
Decision to Pivot	2s			
Secondary Technique	13s			
50–65s: Verification				
Answer Testing	10s			
Final Check	5s			
Total Time	65s			

Decision Gate Analysis:

- ☐ **Clear? (15s)**: Domain and technique identified correctly?
  - YES → Proceeded to execution
  - NO → Moved to pivot strategy
- ☐ **Solved? (35s)**: Made meaningful progress with domain-specific technique?
  - YES → Moved to answer verification
  - NO → Moved to pivot strategy
- ☐ **Fallback Worked? (50s)**: Pivot technique providing clear progress and manageable calculation?
  - YES → Moved to answer verification
  - NO → Moved to answer elimination
- ☐ **Verified? (65s)**: Answer verification successful during testing phase?
  - YES → Proceeded to next problem
  - NO → Moved to answer elimination
- ☐ **Confident? (65s)**: Answer appears solid and constraints are satisfied?
  - YES → Proceeded to next problem
  - NO → Entered red flags protocols

WHO/WHEN Pattern Recognition Assessment

Pattern Recognition Metrics:

- **WHO Identification** (Problem type): ____ / 10 correct
- **WHERE Recognition** (Pattern cues): ____ / 10 spotted
- **WHEN Application** (Timing decision): ____ / 10 optimal

For Each Problem Attempted:

#	WHO (Type)	WHEN (Cue/Conditions)	WHEN (Applied)	Success?

Pattern Library Update:

- ☐ New WHO/WHEN combinations discovered
- ☐ Cue recognition time improved
- ☐ False pattern matches identified and corrected

Pattern Library Entry — Example

New Pattern Found: Coordinate Geometry Fallback
Recognition Cue: Synthetic geometry seems messy; angle chasing stalls; loci/lengths/areas with algebraic relationships; need robust, algorithmic path.
Execution Steps:

1. Place a convenient coordinate frame (use symmetry; set one key point at origin or on axes).
2. Assign variables to unknown points; encode given constraints as equations.
3. Compute with vector/dot product, distance, slope, or area formulas.
4. Solve and back-interpret geometrically; verify conditions.

Time Saved: 60–120 s vs. prolonged synthetic attempts (especially 11–20).
Recognition Latency (s): 5–10 (pivot when stuck by 30–60 s per protocol)
Observed Success Rate: approximately 90% as universal fallback on geometry, per Quick Reference.
Micro-Examples:

- Point-to-line distance: from (x_0, y_0) to $ax + by + c = 0$, $d = \dfrac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- Triangle area by coordinates: $A = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
- Right triangle with altitude to hypotenuse: set vertices at $(0, 0)$, $(c, 0)$, $(0, b)$ to exploit orthogonality.

Common Trap: Over-parameterizing (too many variables); avoid by using symmetry and minimal coordinates.

Pattern Library Entry — Example

New Pattern Found: Completing the Square
Recognition Cue: Single-variable quadratics; expressions of the form $ax^2 + bx + c$; sum of two squares with linear terms; circle/vertex form in coordinate geometry.
Execution Steps:

1. If $a \neq 1$, factor: $ax^2 + bx + c = a\big(x^2 + \frac{b}{a}x\big) + c$.
2. Complete the square: $x^2 + px = (x + \frac{p}{2})^2 - (\frac{p}{2})^2$.
3. Rewrite to vertex/center form; read off minima/maxima or geometric parameters.
4. Solve, optimize, or identify locus as required.

Time Saved: 20–60 s vs. quadratic formula/calculus or ad-hoc geometry.
Recognition Latency (s): 2–4
Observed Success Rate: 90–95% on 1–10; 80–90% on 11–15 when applicable
Micro-Examples:

- Factor/optimize: $x^2 - 6x + 5 = (x - 3)^2 - 4$; minimum -4 at $x = 3$.
- Solve: $x^2 - 6x + 5 = 0 \Rightarrow (x - 3)^2 = 4 \Rightarrow x = 1, 5$.
- Circle: $x^2 + y^2 - 4x + 6y = 12 \Rightarrow (x - 2)^2 + (y + 3)^2 = 25$; center $(2, -3)$, radius 5.

Common Trap: Forgetting to factor a first when $a \neq 1$, or sign errors when moving constants.

7.2 Technique Selection and Conversion Analysis

Technique Application Tracking (Aligned with Companion Documents)

High-Probability Techniques Used (Reference: Conversion Techniques Document):

Technique	Applied	Success	Notes
T1: Algebraic Substitution	__/__	__%	
T2: Coordinate Geometry	__/__	__%	
T3: Case Analysis	__/__	__%	
T4: Symmetry Exploitation	__/__	__%	
T5: SFFT	__/__	__%	
T6: Pigeonhole Principle	__/__	__%	
T7: AM-GM Inequality	__/__	__%	
T8: Completing the Square	__/__	__%	
T9: Base/Digit Conversions	__/__	__%	
T10: Angle Bisector Theorem	__/__	__%	

Verification Level Applied (Reference: Complete Verification Methods Document):

- ☐ L1: Instant Gut Check (1–3s) - Used: ____ times
- ☐ L2: Minimal Sanity Check (5–10s) - Used: ____ times
- ☐ L3: Targeted Spot Check (15–30s) - Used: ____ times
- ☐ L4: Standard Verification (30–60s) - Used: ____ times
- ☐ Higher levels (L5–L8) - Used: ____ times

Conversion Technique Performance:

- Primary technique success rate: ____ / ____ = ___%
- Pivot technique success rate: ____ / ____ = ___%
- Coordinate geometry fallback success: ____ / ____ = ___%

Domain Lane Performance Metrics

Domain	Problems	Correct	Avg Time	Success Rate
ALG (Algebra)				%
GEO (Geometry)				%
CNT (Counting)				%
NT (Number Theory)				%
PROB (Probability)				%

Domain-Specific Issues Identified:

- ☐ Algebra: Sign errors, domain restrictions, extraneous solutions
- ☐ Geometry: Configuration misidentification, scaling errors
- ☐ Counting: Overcounting, order confusion, fence-post errors
- ☐ Number Theory: Modular arithmetic errors, FLT misapplication
- ☐ Probability: Independence assumptions, complement errors

7.3 Error Pattern Deep Dive

Comprehensive Error Analysis

Error Classification Matrix:

Error Type	Frequency	Problems	Time Lost	Remediation Strategy
Conceptual Errors				
Misunderstood problem	__/__			Re-read training
Wrong technique chosen	__/__			Pattern practice
Missing constraint	__/__			Checklist review
Computational Errors				
Arithmetic mistakes	__/__			Mental math drills
Algebraic manipulation	__/__			Step-by-step practice
Sign/notation errors	__/__			Verification habits
Reading Errors				
Misread question	__/__			Underline key words
Missed given info	__/__			Systematic scanning
Wrong answer format	__/__			Answer check protocol
Strategic Errors				
Time misallocation	__/__			Timeline discipline
No pivot when stuck	__/__			30s pivot rule
Skipped verification	__/__			Build habit

Trap Pattern Recognition:

- ☐ **Trap Type Encountered**: _____
- ☐ **Why It Worked**: _____
- ☐ **Warning Signs Missed**: _____
- ☐ **Future Detection Strategy**: _____

Error Correlation Analysis:

- ☐ Errors cluster in specific domains? Y/N → Which: _____
- ☐ Errors increase with problem difficulty? Y/N → Pattern: _____
- ☐ Errors correlate with time pressure? Y/N → Threshold: _____
- ☐ Errors follow fatigue pattern? Y/N → Onset time: _____

7.4 Improvement Action Items

Timeline-Based Improvement Plan

Classification Phase Improvements (0–15s):

- ☐ Domain scan speed: Current ___s → Target 3s
- ☐ WHO/WHEN recognition: Current ___s → Target 3s
- ☐ Answer intelligence: Current ___s → Target 4s
- ☐ Practice missed patterns: _____

Execution Phase Improvements (15–35s):

- ☐ Domain lane entry speed: Current ___s → Target 5s
- ☐ Primary technique mastery needed: _____
- ☐ Mental math drills required: _____
- ☐ Reference Conversion Techniques document sections: T___, T___, T___

Pivot/Verification Improvements (35–65s):

- ☐ Pivot decision speed: Current ___s → Target 2s
- ☐ Answer testing efficiency: Current ___s → Target 10s
- ☐ Trap patterns to review: _____
- ☐ Verification level adjustment needed: L___ → L___

Weekly Practice Schedule:

- ☐ Monday/Wednesday/Friday: 10 problems with full timeline tracking
- ☐ Tuesday/Thursday: Pattern recognition drills (3s each)
- ☐ Saturday: Full practice test with analysis

☐ Sunday: Review and pattern library update

Red Flag Indicators (Immediate Action Required):

☐ Classification accuracy drops below 70% → Review Section 1
☐ Same error type 3+ times → Targeted drill on that error type
☐ Average time exceeds targets by 30+ seconds → Timeline discipline reset
☐ Success rate plateaus for 2 weeks → Strategy overhaul needed
☐ Fatigue onset before 45 minutes → Stamina building required
☐ Pivot success rate below 50% → Practice fallback techniques
☐ Verification catching $< 50\%$ of errors → Upgrade verification level

Quantitative Improvement Metrics

Current Performance Baseline:

- Classification accuracy: ____ / 10 = ___%
- Execution success rate: ____ / ____ = ___%
- Average time per problem (1–10): ____ seconds
- Average time per problem (11–20): ____ seconds
- Decision gate accuracy: ____ / 4 gates = ___%

Fatigue & Stamina Metrics:

- Performance by position: Early (1–7): ___%, Mid (8–14): ___%, Late (15–20): ___%
- Energy level at 30 min: High/Medium/Low
- Energy level at 60 min: High/Medium/Low
- Energy level at 90 min: High/Medium/Low
- Fatigue-error correlation: Strong/Moderate/Weak/None
- Optimal break strategy identified: _____

Target Performance (Next Week):

- Classification accuracy: $\geq 90\%$
- Execution success rate: $\geq 85\%$
- Average time reduction: 15s (problems 1–10), 30s (problems 11–20)
- Decision gate accuracy: $4/4 = 100\%$

Long-term Goals (4 weeks):

- Master all 10 high-probability techniques (T1–T10)
- Achieve 85%+ accuracy on problems 1–15
- Complete problems 1–10 in $\leq$ 90s average
- Successfully apply verification levels per companion guide

7.5 Personal Pattern Library Update

Enhanced Pattern Library Entry Template (WHO/WHEN Framework)

Pattern Identification:

- **Pattern Name**: [Descriptive name]
- **WHO** (Problem types): [List applicable problem categories]
- **WHERE** (Recognition cues): [Specific keywords, structures, configurations]
- **WHEN** (Application conditions): [Constraints, time windows, triggers]

Execution Protocol:

- **Primary Technique**: [Reference: T___ from Conversion Techniques]
- **Fallback Technique**: [Reference: T___ from Conversion Techniques]
- **Timeline Integration**: [Which phase: Classification/Execution/Pivot/Verify]
- **Mental Math Components**: [List any mental calculations]

Performance Metrics:

- **Recognition Latency**: ____ seconds (target: $\leq$ 3s)

- **Application Success Rate**: ____ / ____ = ___%
- **Time Saved vs. Brute Force**: ____ seconds
- **Verification Level Used**: L___ (per Verification Methods guide)

Common Variations:

- **Trap Version**: [How it might appear as a trap]
- **Complex Version**: [How it appears in problems 16–20]
- **Cross-Domain Version**: [Mixed with other domains]

Pattern Recognition Speed Tracking

Pattern Type	Current	Target	Best	Notes
Algebraic substitution cues	__s	2s		
SFFT structure	__s	2s		
Coordinate setup triggers	__s	3s		
AM-GM opportunities	__s	2s		
Power of Point configs	__s	3s		
Counting complement	__s	2s		
Vieta's application	__s	2s		
Case analysis need	__s	3s		

7.6 Post-Competition Analysis Summary

Comprehensive Performance Review

Overall Timeline Adherence:

Problem Band	Target Time	Actual Avg	Within Target?
1–5	65–120s	___s	Y/N
6–10	120–180s	___s	Y/N
11–15	180–240s	___s	Y/N
16–20	240–300s	___s	Y/N

Classification System Effectiveness:

- 10-second classification success rate: ____ / ____ = ___%
- WHO/WHEN framework utility: Excellent / Good / Fair / Poor
- Decision gates navigated correctly: ____ / ____ gates
- Timeline discipline maintained: Always / Usually / Sometimes / Rarely

Companion Document Integration:

- Conversion Techniques applied successfully: ____ / ____
- Verification Methods used appropriately: ____ / ____
- Cross-reference speed (finding techniques): Fast / Moderate / Slow
- Technique numbering system helpful: Yes / No

Key Successes to Replicate:

1. _____
2. _____
3. _____

Priority Areas for Improvement:

1. _____
2. _____
3. _____

Next Session Focus:

☐ Primary: _____

☐ Secondary: _____

☐ Review sections from guides: _____

Cross-Competition Performance Comparison:

Metric	AMC12	COMC	Transfer Opportunity
Problems 1–5 accuracy	___%	___%	
Problems 6–10 accuracy	___%	___%	
Problems 11–15 accuracy	___%	___%	
Average time/problem	___s	___s	
Domain strengths			
Domain weaknesses			
Best techniques			

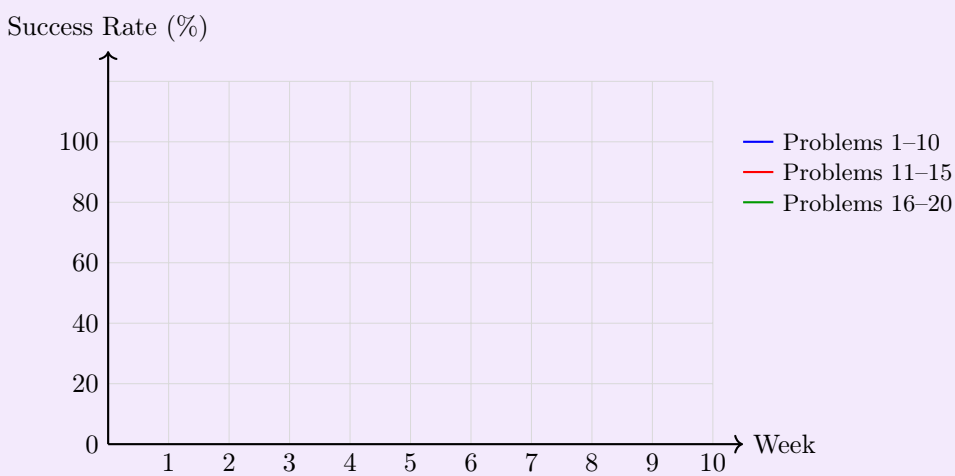
Competition-Specific Adjustments:

☐ AMC12 emphasis: Speed on 1–15, strategic guessing on 16–25
☐ COMC emphasis: Deep exploration, partial credit optimization
☐ Technique modifications needed: _____
☐ Timeline adjustments required: _____

7.7 Progress Visualization Guide

Performance Tracking Visualization

Weekly Progress Graph Template:
Draw your progress on this grid (or use spreadsheet/graph paper):



Key Metrics to Track:

1. **Accuracy Curves**: Plot weekly success rates for each problem band
2. **Speed Progression**: Average time per problem (separate line)
3. **Milestone Markers**: Mark when you achieve key goals
4. **Technique Mastery**: Track # of techniques at 80%+ success rate

Milestone Progression Checklist:

☐ **Week 1**: Baseline established, timeline memorized
☐ **Week 2**: 80%+ on problems 1–5 consistently
☐ **Week 3**: Classification under 10s for all problems
☐ **Week 4**: 85%+ on problems 1–10
☐ **Week 6**: 70%+ on problems 11–15
☐ **Week 8**: All Tier 1 techniques mastered
☐ **Week 10**: Competition-ready performance achieved

Performance Plateau Indicators:

- Same accuracy for 2+ weeks → Change practice approach
- Time not improving → Focus on pattern recognition speed
- Repeated error types → Targeted remediation needed
- Energy dropping → Adjust practice schedule/intensity

Suggested Digital Tools:

- Spreadsheet with automatic graphing (Google Sheets/Excel)
- Practice log app with timer and accuracy tracking
- Pattern library database (Notion, Obsidian, or similar)
- Video review of problem-solving sessions for self-analysis

8 Early/Mid-Band Mastery Checklist

Prerequisites Before Using This Checklist

- ☐ Familiar with companion documents:
  - ☐ AMC12_COMC_Conversion_Techniques.tex (49 techniques)
  - ☐ AMC12_COMC_Complete_Verification_Methods.tex (8 levels)
- ☐ Completed at least 50 practice problems in each band (1–10 and 11–20)
- ☐ Understand the WHO/WHEN framework for pattern recognition
- ☐ Familiar with domain tagging system (ALG/GEO/CNT/NT/PROB)
- ☐ Have practiced with timer for classification phases

Rapid Classification Mastery (Problems 1–10)

- ☐ 15-Second Classification Method mastered:
  - ☐ Domain Scan (0–4s): Visual scan protocol with domain tagging
  - ☐ Pattern Match (4–10s): WHO/WHEN framework applied
  - ☐ Answer Intelligence (10–15s): Answer-choice analysis automated
- ☐ Timeline execution consistent:
  - ☐ 0–15s: Classification complete with decision gates
  - ☐ 15–35s: Domain lanes execution
  - ☐ 35–50s: Pivot protocol automatic
  - ☐ 50–65s: Answer testing proficient
  - ☐ 65–120s: Emergency bailout criteria clear
- ☐ Tier 1 conversions (80–95% success rate) mastered:
  - ☐ Algebraic Substitution; Vieta's Relations and Factoring
  - ☐ Similar Triangles, Homothety, and Angle Chasing
  - ☐ Coordinate Geometry (distance, area, loci)
  - ☐ Strategic Bounds/Inequalities (incl. AM-GM)
  - ☐ Completing the Square
- ☐ Pythagorean triples/special right triangles recognized
- ☐ Mental math shortcuts automated:
  - ☐ Squares 11–20 memorized (121, 144, 169, 196, 225, 256, 289, 324, 361, 400)
  - ☐ Common factorials memorized (up to $10! = 3{,}628{,}800$)
  - ☐ Powers of 2 memorized (up to $2^{16} = 65{,}536$)
  - ☐ Quick fraction-decimal conversions ($\frac{1}{6} = 0.1\overline{6}$, $\frac{1}{7} \approx 0.143$, etc.)
  - ☐ Common roots approximated ($\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$)
- ☐ Trap patterns recognized instantly:
  - ☐ "Too easy" answers in problems 8–12 (usually missing a constraint)
  - ☐ Missing domain restrictions (denominators $\neq 0$, logs require positive arguments)
  - ☐ Extraneous solutions from squaring both sides or clearing denominators
  - ☐ Off-by-one counting errors (fence-post problems, inclusive vs. exclusive)
  - ☐ Sign errors in absolute value problems
  - ☐ Circular arrangement overcounting (forgetting to divide by n for rotations)
  - ☐ Assuming perpendicularity or special angles without justification
- ☐ Consistent time: 60–120 seconds (1–5), 120–180 seconds (6–10) with 90%+ success rate

Advanced Classification and Execution Mastery (Problems 11–20)

- ☐ Extended timeline execution proficient:
  - ☐ 11–15: 180–240s total with extended execution phase
  - ☐ 16–20: 240–300s with complex work phases
  - ☐ Decision gates at 180s and 240s automated

- ☐ Domain-specific lanes mastered:
  - ☐ Algebra Lane: SFFT, hidden quadratics, u-substitution
  - ☐ Geometry Lane: Power of Point, angle bisector theorem, coordinates
  - ☐ Counting Lane: Inclusion-exclusion, stars and bars, symmetry
  - ☐ Number Theory Lane: FLT, CRT, digit rules
  - ☐ Probability Lane: Linearity of expectation, indicators
- ☐ Tier 2 conversions (40–70% success rate) practiced:
  - ☐ Modular Arithmetic (residues/cycles); Chinese Remainder Theorem (Optional/Advanced)
  - ☐ Power of a Point; Cyclic Quadrilaterals; Ptolemy (Optional/Advanced)
  - ☐ Stewart's Theorem; Ceva/Menelaus (Optional/Advanced)
  - ☐ Invariants/Monovariants; Recurrences/Characteristic Roots
  - ☐ Trigonometric Substitution
  - ☐ Generating Functions (Optional/Advanced)
- ☐ 30-second pivot protocol automatic with fallback hierarchy
- ☐ Conversion escalation path memorized
- ☐ Consistent time: 180–240s (11–15), 240–300s (16–20) with 75%+ success rate

Competition-Ready Integration Checklist

- ☐ Rapid Classification System fully internalized:
  - ☐ 10-second classification automatic for all problems
  - ☐ WHO/WHEN framework applied instinctively
  - ☐ Decision gates (Clear? Solved? Fallback Worked? Verified? Confident?) memorized
- ☐ Pattern library linked to conversion techniques (49 numbered techniques: 25 high-probability + 24 medium-probability)
- ☐ Success probability thresholds memorized for each technique
- ☐ Time allocation aligned with classification timeline:
  - ☐ Classification phase always 0–15s
  - ☐ Execution/pivot/verify phases scale with problem difficulty
  - ☐ Hard stops enforced: 2 min (1–10), 3.5 min (11–15), 4.5 min (16–20)
- ☐ Verification levels selected per band:
  - ☐ L2 Minimal Sanity Check (5–10s) for problems 1–10
  - ☐ L3 Targeted Spot Check (15–30s) for problems 11–20
  - ☐ Higher levels (L4–L8) for problems 19–20 when time permits
- ☐ Universal fallbacks mastered:
  - ☐ Geometry: Coordinate conversion (65–90% success)
  - ☐ Algebra: u-substitution (75–85% success)
  - ☐ Optimization: Strategic bounds (70–80% success)
  - ☐ Counting: Case analysis (70–75% success)
- ☐ 85%+ accuracy on problems 1–10 using rapid classification
- ☐ 75%+ accuracy on problems 11–20 using extended timeline
- ☐ Average time saved: 40%+ with systematic approach
- ☐ Can execute classification → execution → verification in target times

Progress Tracking Metrics

- ☐ Baseline assessment completed:
  - ☐ Initial AMC12 score established: ___/150 (___/25 problems)
  - ☐ Time per problem band documented: 1–10: ___s avg, 11–20: ___s avg
  - ☐ Strongest domains identified: _____
  - ☐ Weakest domains identified: _____
- ☐ Weekly practice metrics:

- ☐ 20+ problems completed per week (minimum)
- ☐ Classification accuracy tracked: Week 1: ___%, Week 2: ___%, Week 3: ___%
- ☐ Average time improving: Target 10% reduction per month
- ☐ Success rate by band: 1–10: ___%, 11–15: ___%, 16–20: ___%

☐ Common mistake patterns identified and addressed:

- ☐ Most frequent classification errors: _____
- ☐ Most common technique failures: _____
- ☐ Recurring trap patterns fallen for: _____
- ☐ Remediation plan created and executed

☐ Milestone achievements:

- ☐ First perfect 1–10 run: Date: _____
- ☐ Consistent 85%+ on problems 1–10: Date: _____
- ☐ Consistent 75%+ on problems 11–15: Date: _____
- ☐ Successfully attempted problem 16+: Date: _____

☐ Competition readiness indicators:

- ☐ Can complete problems 1–10 in under 25 minutes consistently
- ☐ Classification phase automatic (no conscious thought required)
- ☐ Pivot decisions made confidently at 35s mark
- ☐ Mock exam scores: Test 1: ___/150, Test 2: ___/150, Test 3: ___/150

9 Appendix: Quick Reference Tables

Note: Items marked "Optional/Advanced" are rarely required for AMC12 problems 1–20.

Important: No calculators are permitted on the AMC12 competition. All calculations must be done by hand.

Cross-Reference: For complete details on all 49 conversion techniques with worked examples and practice problems, consult `AMC12_COMC_Conversion_Techniques.tex`. The techniques are numbered in that document for easy reference.

9.1 Rapid Classification Timeline Success Rates

Problem Band	Classification (0–15s)	Execution (15–35s)	Pivot (35–50s)	Verification (50–65s)	Overall Success
1–5	95%	92–95%	85%	90%	95%+
6–10	90%	85–92%	80%	85%	90–92%
11–15	80%	75–85%	70%	75%	80–85%
16–20	70%	60–75%	60%	65%	70–75%

Key: Success rates assume proper execution of the Rapid Classification System timeline.
Note: Tier 1 conversions apply to Execution phase; Tier 2 conversions apply to Pivot phase.

9.2 Time Allocation Summary with Classification Timeline

Problem #	Target Total	Classification	Hard Stop	Verification Level
1–5	65–120s	0–15s	120s	L2 (5–10s)
6–10	120–180s	0–15s	120s	L2 (5–10s)
11–15	180–240s	0–15s	210s	L3 (15–30s)
16–20	240–300s	0–15s	270s	L3+ (30s+)

Decision Gates: Clear? (15s) → Solved? (35s) → Fallback Worked? (50s) → Verified? (65s) → Confident? (65s)

9.3 15-Second Universal Fallback Methods (35-Second Pivot Protocol)

Domain	Primary Fallback	Secondary Fallback	Success Rate
Algebra	u-substitution	Test answer choices	75–85%
Geometry	Coordinate geometry	Special positions	65–90%
Counting	Case analysis	Small cases pattern	70–75%
Number Theory	Modular arithmetic	Check small primes	60–70%
Probability	Linearity of expectation	Complement method	65–75%
Optimization	Strategic bounds	Test extremes	70–80%

Trigger: Use fallbacks at 35s if no progress in execution phase (15–35s).

9.4 Domain Lanes Quick Reference

Domain	Primary Techniques	Key Cues	Success Rate
Algebra	SFFT (Simon's Favorite)	Bilinear $xy + ax + by$	85%
	Vieta's Relations	Sum/product of roots	90%
	u-substitution	Repeated chunks	85%
Geometry	Power of a Point	Chords/tangents	80%
	Angle Bisector Theorem	Angle bisector present	85%
	Coordinate Geometry	Complex figure	75%
Counting	Inclusion-Exclusion	"At least" language	80%
	Stars and Bars	Identical objects	85%
	Casework by symmetry	Natural partitions	75%
Number Theory	Fermat's Little Theorem	Large exponents, prime	85%
	Chinese Remainder	Multiple moduli	70%
	Digit Rules	Last digits, digit sums	80%
Probability	Linearity of Expectation	Expected counts	85%
	Complement Method	"At least" probability	90%

Common Domain Pitfalls to Avoid:

Domain	Common Pitfalls
Algebra	Forgetting negative roots; dividing by expressions that could be zero
Geometry	Missing cases in configuration; assuming diagrams are to scale
Counting	Double-counting or missing cases; confusing permutations and combinations
Number Theory	Forgetting gcd conditions; misapplying modular arithmetic rules
Probability	Assuming independence when not given; forgetting to normalize

9.5 Mental Math Must-Know Values

Squares	Value	Constants	Value
11^2	121	π	≈ 3.14159
12^2	144	e	≈ 2.718
13^2	169	$\sqrt{2}$	≈ 1.414
14^2	196	$\sqrt{3}$	≈ 1.732
15^2	225	$\sqrt{5}$	≈ 2.236
16^2	256	ϕ	≈ 1.618
17^2	289	$\ln 2$	≈ 0.693
18^2	324	$\ln 10$	≈ 2.303
19^2	361		
20^2	400		

Factorials	Value	Powers of 2	Value
0!	1	2^5	32
1!	1	2^6	64
2!	2	2^7	128
3!	6	2^8	256
4!	24	2^9	512
5!	120	2^{10}	1024
6!	720	2^{11}	2048
7!	5040	2^{12}	4096
8!	40320	2^{13}	8192
9!	362880	2^{14}	16384
10!	3628800	2^{15}	32768
11!	39916800	2^{16}	65536

9.6 Verification Levels Quick Reference

Level	Time	When to Use	Actions
L1: Gut Check	1–3s	Quick mental scan	Reasonableness only
L2: Minimal Sanity	5–10s	Problems 1–10	Units, bounds, special cases
L3: Targeted Spot	15–30s	Problems 11–20	Test key values, check work
L4: Standard	30–60s	When time permits	Full re-calculation
L5: Thorough	1–3m	Problems 19–20	Alternative method
L6: Multi-Method	3–5m	High-value problems	2+ independent methods
L7: Complete	5–10m	Final problems	Mathematical rigor
L8: Formal Proof	10m+	Post-competition	Full proof

Key Principle: Match verification level to problem difficulty and available time.

9.7 Common AMC12 Formulas Quick Reference

The following comprehensive formula reference covers all major topics appearing in AMC12 problems 1–20. Formulas are organized by domain for quick access during competition.

Topic	**Formula**
Factorials	$0! = 1$, $n! = n \cdot (n-1)!$, $\frac{n!}{(n-k)!} = n(n-1)\cdots(n-k+1)$
Permutations	$P(n,r) = \dfrac{n!}{(n-r)!}$
Combinations	$C(n,r) = \binom{n}{r} = \dfrac{n!}{r!(n-r)!}$
Quadratic Formula	$x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$
Vieta's Formulas	For $ax^2+bx+c=0$ roots r_1, r_2: $r_1+r_2 = -\frac{b}{a}$, $r_1r_2 = \frac{c}{a}$
Sum of Arithmetic Series	$S_n = \frac{n(a_1+a_n)}{2}$
Sum of Geometric Series	$S_n = a\frac{1-r^n}{1-r}$ (for $r \neq 1$)
Infinite Geometric Series	$S_\infty = \dfrac{a}{1-r}$ (for $\lvert r\rvert < 1$)
Binomial Expansion	$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$
Law of Sines	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
Law of Cosines	$c^2 = a^2+b^2-2ab\cos C$
Triangle area (sine)	$A = \frac{1}{2}ab\sin C$
Heron's Formula	$A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$
Sequences	**Formula**
Arithmetic nth term	$a_n = a_1 + (n-1)d$
Geometric nth term	$a_n = a_1 \cdot r^{n-1}$
Sum of first n integers	$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$
Sum of first n squares	$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$
Sum of first n cubes	$\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$
Complex Numbers	**Formula**
Modulus	$\lvert a+bi\rvert = \sqrt{a^2+b^2}$
Multiplication	$(a+bi)(c+di) = (ac-bd)+(ad+bc)i$
Division	$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$
De Moivre's Theorem (Optional/Advanced)	$(r\text{cis}\theta)^n = r^n\text{cis}(n\theta)$
cis definition	$\text{cis}\,\theta = \cos\theta + i\sin\theta$
Euler's Formula	$e^{i\theta} = \cos\theta + i\sin\theta$
Trigonometric Identities	**Formula**
Pythagorean Identity	$\sin^2\theta + \cos^2\theta = 1$
Double Angle (sine)	$\sin(2\theta) = 2\sin\theta\cos\theta$
Double Angle (cosine)	$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$
Half Angle	$\sin^2\left(\frac{\theta}{2}\right) = \frac{1-\cos\theta}{2}$, $\cos^2\left(\frac{\theta}{2}\right) = \frac{1+\cos\theta}{2}$
Sum Formulas (Optional/Advanced)	$\sin(A\pm B) = \sin A\cos B \pm \cos A\sin B$ $\cos(A\pm B) = \cos A\cos B \mp \sin A\sin B$
Tangent Identity	$\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\tan^2\theta + 1 = \sec^2\theta$
Logarithm Properties	**Formula**
Product Rule	$\log_b(xy) = \log_b(x) + \log_b(y)$
Quotient Rule	$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$
Power Rule	$\log_b(x^n) = n\log_b(x)$
Change of Base	$\log_b(x) = \frac{\log_a(x)}{\log_a(b)}$
Special Values	$\log_b(1) = 0$, $\log_b(b) = 1$, $\log_b(b^n) = n$
Natural Log Properties	$\ln(e) = 1$, $e^{\ln x} = x$, $\ln(e^x) = x$
Floor/Ceiling	**Property**
Floor definition	$\lfloor x\rfloor = \max\{n \in \mathbb{Z} : n \leq x\}$
Ceiling definition	$\lceil x\rceil = \min\{n \in \mathbb{Z} : n \geq x\}$
Relationship	$\lceil x\rceil = -\lfloor -x\rfloor$
Integer addition	$\lfloor x+n\rfloor = \lfloor x\rfloor + n$ for $n \in \mathbb{Z}$
Fractional part	$\{x\} = x - \lfloor x\rfloor$
Coordinate Geometry	**Formula**
Shoelace (polygon area)	$A = \frac{1}{2}\left\lvert\sum_{i=1}^{n} x_iy_{i+1} - y_ix_{i+1}\right\rvert$, with $(x_{n+1}, y_{n+1}) = (x_1, y_1)$
Point-to-line distance	$d = \dfrac{\lvert ax_0+by_0+c\rvert}{\sqrt{a^2+b^2}}$ from (x_0, y_0) to $ax+by+c=0$

9.8 Rapid Classification System Quick Reference Card

Competition Day Quick Reference

Master Timeline (ALL Problems):
- **0–15s**: Classification (Domain + Pattern + Answers)
- **15–35s**: Execution (Domain lanes)
- **35–50s**: Pivot (if stuck)
- **50–65s**: Verification (answer test)
- **65–120s**: Emergency bailout

15-Second Classification Method:
1. **0–4s**: Domain Scan (ALG/GEO/CNT/NT/PROB)
2. **4–10s**: Pattern Match (WHO/WHEN)
3. **10–15s**: Answer Intelligence

Decision Gates:
- **15s**: Clear?
  - YES → Execute domain-specific technique
  - NO → Move to Pivot Strategy
- **35s**: Solved?
  - YES → Move to Answer Verification
  - NO → Move to Pivot Strategy
- **50s**: Fallback Worked?
  - YES → Move to Answer Verification
  - NO → Move to Answer Elimination
- **65s**: Verified?
  - YES → Proceed directly to Next Problem
  - NO → Move to Answer Elimination
- **65s**: Confident?
  - YES → Proceed to Next Problem
  - NO → Enter Red Flags protocols

Problems 1–10 Protocol:
1. **0–15s**: Rapid classification
2. **15–35s**: Execute Tier 1 conversion
3. **35–50s**: Pivot if needed
4. **50–65s**: L2 Verification (5–10s)
5. **65–120s**: Emergency bailout if needed

Problems 11–20 Protocol:
1. **0–15s**: Rapid classification
2. **15–35s**: Execute domain lane
3. **35–50s**: Pivot to Tier 2 if stuck
4. **50–65s**: L3 Verification (15–30s)
5. **Extended**: 180–300s total allowed
6. **For 19–20**: Escalate verification if time permits

Universal Fallbacks (35s Pivot):
- **Geometry**: Coordinates (65–90% success)
- **Algebra**: u-substitution (75–85% success)
- **Optimization**: Strategic bounds (70–80% success)
- **Counting**: Case analysis (70–75% success)

Domain Lane Cues (15–35s):
- **ALG**: SFFT, Vieta's, complete square
- **GEO**: PoP, ABT, coordinates
- **CNT**: Inc-Exc, stars & bars
- **NT**: FLT, CRT, digit rules
- **PROB**: Linearity, indicators

Answer-Testing Triggers (50–65s):
- Algebra with verified monotonicity → test median-magnitude as scale check
- Geometry radicals → check Pythagorean
- Small sample space → enumerate cases

Success Formula: Classification (10s) + Execution (20s) + Verification (15s) = Target Time

— End of AMC12 Early/Mid-Band Mastery Guide (INTEGRATED EDITION) —

Master the patterns. Apply the conversions. Maintain the discipline. Achieve the score.